Introduction to Thyroid

"Every cell in the body has receptor sites for thyroid hormones. Thyroid hormones are responsible for the most basic and fundamental aspect of physiology, the basal metabolic rate. Lack of ideal thyroid hormone leads to global decline in cellular function of all bodily systems. The thyroid is the central gear in the complex web of metabolism and is extremely sensitive to minor imbalances in other areas of physiology. An astute clinician should always ask what else is going wrong, as a result and cause, when recognizing the signs and symptoms of a thyroid imbalance."

Datis Kharrazian, DC, DHSc, FACN, DACBN, DACNB, CNS

Disorders for thyroid function are very prevalent in the United States population and continue to increase every year. Thyroid hormones (especially Synthroid®) have been on the top 50 most-prescribed medications for decades. We are all prone to thyroid disorders and nodules as we get older. Most thyroid conditions are not diagnosed in the early stages and lead to loss of function, resources, and quality of life. Hypothyroidism is the most common cause of thyroid dysfunction, followed by hyperthyroidism goiter and thyroid cancer.

Please see registered trademark ownership in Appendix 1.

Prevalence of Thyroid Disorders in the U.S. in 2002	
Hypothyroidism	10,620,000
Hyperthyroidism	1,435,000

Source: Aoki Y, Belin RM, Clickner R, Jeffries R, Phillips L, Mahaffey KR. Serum TSH and total T4 in the United States population and their association with participant characteristics: National Health and Nutrition Examination Survey (NHANES 1999-2002). Thyroid. 2007 Dec;17:1211-1223.

Incidence of Thyroid Disorders in the U.S. in 2014	
Thyroid Cancer	62,980
Deaths Due to Thyroid Cancer	1,890

Source: American Cancer Society. Thyroid Cancer. http://www.cancer.org/cancer/thyroidcancer/detailedguide/thyroid-cancer-key-statistics.

Thyroid physiology is very sensitive to imbalances in physiological systems. The thyroid gland is the most common site for autoimmune attack and oftentimes the first sign of the development of an autoimmune disease. Hashimoto's disease is considered the most prevalent autoimmune disorder in the United States.[1] Hashimoto's disease has been found to be the mechanism for hypothyroidism in 90% of the cases in the United States.[2] One study has found that up to 7%-8% of the U.S. population have antibodies against their thyroid.[1] Thyroid antibodies have indeed been found to be a marker for future thyroid disease.[3] Sadly, these patients' thyroids are slowly destroyed year by year until they have finally lost enough of their thyroid tissue to be classified as hypothyroid. As they progressively lose thyroid tissue, they also decline in their basal metabolic rate and start having gradual decline in their energy, mental function, gastrointestinal motility, mood, etc. Even more disheartening, when they are finally diagnosed, their treatment options are limited solely to thyroid hormone replacement. Nothing is done for their autoimmune disease, and they progressively decline or feel no improvement with management. The thyroid gland is not only vulnerable to autoimmune disease. Immunological activity has also been shown to significantly disrupt the hypothalamic-pituitary-thyroid axis, peripheral thyroid hormone metabolism, thyroid receptor-site sensitivity, and thyroid receptor-proteomic expressions. Many chronically immune-challenged patients suffer from chronic immune activation (hepatitis C, Lyme disease, an unknown virus, inflammatory bowel disease, autoimmune disease, etc) that leads to elevated cytokines (TNF, IFN, IL-6, etc) and disruption of healthy thyroid physiology. Therefore, thyroid disorders are simply at times a warning light for immune dysfunction. Clinical management directed at restoring immune activity has the greatest promise in supporting these thyroid consequences.

The thyroid gland is also very vulnerable to imbalances of the endocrine system. Hormones, such as estrogens, progesterone, cortisol, and testosterone, have a major influence on thyroid peroxidase enzymes, 5'-deiodinase enzymes, thyroid-binding globulins, as well as thyroid receptor-site sensitivities. Many times, other endocrine imbalances are the culprit in thyroid imbalances, and restoring these imbalances has the greatest promise in supporting thyroid metabolic dysfunction.

The thyroid gland is very vulnerable to environmental disruptors, some known and many unknown. Many of the known environmental disruptors act as goitrogens and compete with iodine uptake. These include mercury, arsenic, nitrates, pesticides, plastic compounds, and ions such as thiocyanates. These compounds may promote goiter activity and disrupt normal thyroid hormone synthesis. Environmental insults in combination

with iodine deficiency, selenium deficiency, underlying immune genotypes, and phenotypes have the greatest potential for thyroid demise.

Thyroid physiology is very vulnerable to cross-reactions with medications. Medications may decrease thyroid hormone secretion, increase secretion, decrease absorption of hormones, decrease thyroid-binding globulin production, increase thyroid-binding globulin production, alter thyroid hormone transport, decrease T4-to-T3 conversion, and alter thyroid receptor-site responsiveness. Unfortunately, the medications that can disrupt thyroid health are commonly prescribed. They include anti-inflammatory agents, antibiotics, antidepressants, diabetic medications, hypertensive medications, pain medication, antacids, and cholesterol-lowering medications. Many of these medications may disrupt thyroid hormone physiology, but never be identified because they may not cause changes in the serum thyroid-stimulating hormone (TSH).

When all of this is taken into clinical context, the astute clinician must consider these variables and consider the fact that the thyroid, due to its vulnerability, may have its demise promoted from other mechanisms, such as autoimmunity, immune-activated cytokines, hormones, environmental compounds, and/or medications. These variables, in conjunction with the necessary steps for thyroid hormone function, can promote physiological thyroid imbalances, not always strict hypothyroidism.

The astute clinician should have a thorough system of evaluation that considers all these variables. These include history-taking skills to identify mechanisms, proper thyroid physical examination to assess for subtle signs, and proper understanding and applications of thyroid serum markers and other diagnostic procedures. Additionally, the astute clinician should have an understanding of clinical applications to each of the mechanisms that exist in order to serve his or her patient.

In this manual, we will first start with a basic review of thyroid physiology and biochemistry. It would be difficult to identify dysfunction without a proper understanding of normal function first. Ten metabolic steps are required for proper thyroid hormone synthesis. These steps include proper serotonergic and dopaminergic activity in the hypothalamus, proper pituitary responses, proper thyroid-peroxidase enzyme activity, proper iodine uptake to the thyroid, proper amino acid synthesis of tyrosine, proper attachment and detachment of thyroid-binding proteins, proper 5'-deiodinase enzyme activity, proper intestinal sulfatase and acetylation enzyme activity, proper nuclear receptor-site sensitivity responses, and proper thyroid receptor -proteomic expressions. Dysfunction in any of these steps can be promoted by cytokines, endocrine hormones, environmental compounds, nutritional deficiencies, autoimmune responses, etc.

The manual will then review how thyroid hormones influence the cells and physiological systems of the body such as the brain, the gastrointestinal tract, bone metabolism, red blood cell metabolism, the immune system, the cardiovascular system, male- and female-hormone steroidogenesis, gallbladder and liver function, body composition and thermogenics, glucose metabolism, lipid and cholesterol metabolism, protein metabolism, etc. An understanding of these systems will help the astute clinician have greater perspective in understanding the spectrum of associated signs, symptoms, and disorders that accompany thyroid demise. These include subtle clinical findings such as hypercholesterolemia, anemia, elevated transaminases, depression, etc.

Thyroid diagnostic tests and procedures will be reviewed. These include serum testing, imaging studies, and biopsy. Serum markers that evaluate thyroid function include: TSH, total T4, total T3, free T4, free T3, TPO Ab, TGB Ab, TSI Ab, reverse T3, thyroglobulin, T3 uptake, and the free thyroxine index. Imaging studies include thyroid ultrasound and radioactive iodine uptake. An understanding of radioiodine uptake will be reviewed and followed by the conditions that cause focal, diffuse, hot, or cold uptakes. The utilization, application, and findings from fine-needle aspiration (FNA) biopsy will be presented. An understanding of proper testing procedures, their risks and benefits, as well as their interpretations are of critical importance for the mismanaged thyroid patient in the healthcare system today.

A step-by-step approach for conducting a thyroid physical examination will be provided as well as a review of common signs of thyroid disorders seen in the skin, eyes, hair, nail beds, personality, cardiovascular system, neurological system, musculoskeletal systems, etc. A step-by-step approach to general inspection, hand inspection, neck inspection, neck palpation, eye examination, motor examination, tibial examination, and neck and thyroid auscultation will be provided. An astute

clinician should be able to identify thyroid dysfunction during an initial examination and history.

A review of medications that alter thyroid metabolism will be provided. The physiological mechanisms that cause disruption will be detailed. The classification of drug types and drug names will be provided. Lastly, a quick form to review these interactions has been formulated and will be reviewed.

An understanding of thyroid goiter, nodules, and masses will be presented. These include mechanisms and clinical findings to help differentially diagnose and properly evaluate these structures if discovered by the patient or during the thyroid examination. A breakdown of thyroid enlargements into three main categories that include diffuse enlargement, multinodular, or a solitary nodule with follow-up clinical differentiation is necessary. Thyroid masses are very common. As many as 60% of people autopsied without a thyroid condition had growths on the thyroid, and 17% had small areas of cancer that were not detected during life. It is important for the astute clinician to identify these masses and have a system for further evaluation in place.

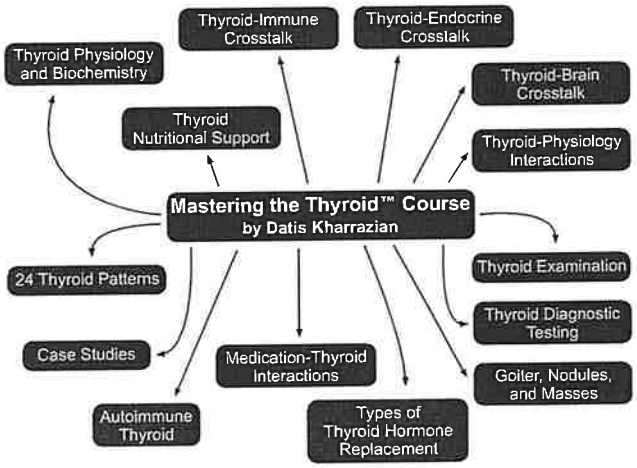
A quick review of the signs, symptoms, clinical presentation, pathophysiology, laboratory findings, diagnostic procedures, and conventional management will be provided for the most common thyroid disorders. These include thyroid metabolic disorders, masses, thyroiditis, cancers, and autoimmune conditions. This section of the manual should be a quick summary for the spectrum of thyroid conditions, as well as provide a summarized resource for the busy clinician.

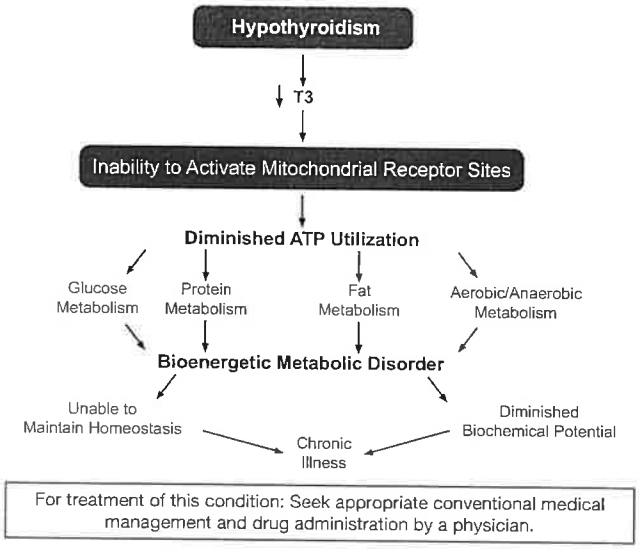
A review of the immune-related mechanisms via cytokines, prostaglandins, lipopolysaccharides, and other immune signaling compounds that disrupt thyroid disorders will be presented. Additionally, a detailed view of Hashimoto's pathophysiology, immunology, and research for triggers is provided. A discussion of TH-1 and TH-2 polar shifts in association with autoimmune related thyroid disorders and its clinical application to immunological testing and nutritional support will be provided and followed by case studies.

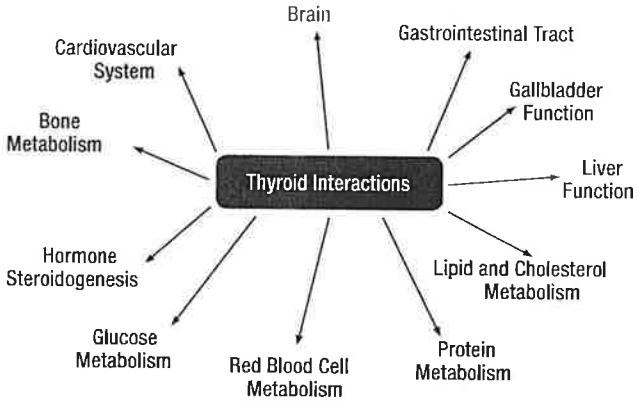
A review of different types and classification of thyroid hormones will be provided. A list of common brand names and their classification as either bioidentical or synthetic will be presented. The classifications of T4 only, T3 only, or T3 and T4 combination hormones are presented. We will provide a history of thyroid hormone development in healthcare that will help the clinician understand the complex political and clinical spectrum of variables that have led to common practices of thyroid hormone management. How different thyroid hormones influence laboratory testing is explained. A section regarding the clinical decision-making process in choosing the proper thyroid hormone specific to the clinical scenario is provided. A conversion chart between different thyroid hormones is included and the section is completed with a discussion on how to determine the meaning of the patient's varied responses to different types of thyroid hormones.

We will provide a review of the literature on nutrients, minerals, and botanicals and how they influence thyroid metabolic support. Additionally, the nutrients will be divided into the mechanisms they support. These include: hypothalamic, pituitary, thyroid hormone synthesis, peripheral hormone synthesis, or receptor-site support. A list of formulations and their applications for nutritional support will be provided.

The manual will conclude with 24 sites of thyroid metabolic defects that start with the hypothalamic-pituitary-axis and end with the receptor sites. A review of the clinical presentation, laboratory markers, and nutritional recommendations will be provided for each pattern. Each one of these patterns can be clearly identified and can relate to imbalances caused by cytokines, medication, nutrient deficiencies, and hormones. The last few sections of the manual will include flow charts, case studies, and references that will be utilized during the seminar presentation.







TOP 50 MOST PRESCRIBED MEDICATIONS

1. Hydrocodone (pain)
2. Lisinopril (hypertension)
3. Simvastatin (high cholesterol)
4. Levothyroxine (hypothyroidism)
5. Amoxicillin (bacterial infection)
6. Azithromycin (bacterial infection)
7. Lipitor® (high cholesterol)
8. Hydrochlorothiazide (edema/hypertension)
9. Alprazolam (anxiety/depression)
10. Atenolol (hypertension)
11. Metformin (type 2 diabetes)
12. Metoprolol Succinate (hypertension)
13. Furosemide, oral (edema/hypertension)
14. Metoprolol tartrate (hypertension)
15. Sertraline (depression)
16. Omeprazole (ulcers/reflux)
17. Zolpidem tartrate (insomnia)
18. Nexium® (ulcers/reflux)
19. Lexapro® (depression)
20. Oxycodone (pain)
21. Singulair® (asthma)
22. Ibuprofen (pain/inflammation)
23. Plavix® (blood clotting)
24. Prednisone, oral (allergies/inflammation)
25. Fluoxetine (depression)
26. Synthroid® (hypothyroidism)
27. Warfarin (blood clotting)
28. Cephalexin (bacterial infection)
29. Lorazepam (anxiety)
30. Clonazepam (epilepsy/anxiety)
31. Citalopram HBr (depression)
32. Tramadol (pain)
33. Gabapentin (epilepsy/pain)
34. Ciprofloxacin HCl (bacterial infection)
35. Propoxyphene-N (pain)
36. Lisinopril (hypertension)
37. Triamterene (edema/hypertension)
38. Amoxicillin (bacterial infection)
39. Cyclobenzaprine (muscle injury/spasm)
40. Prevacid® (ulcers/reflux)
41. Advair® (asthma)
42. Effexor XR® (depression)
43. Trazodone HCl (depression)
44. Fexofenadine (allergy)
45. Fluticasone nasal spray (allergy)
46. Diovan® (hypertension)
47. Paroxetine (depression/anxiety)
48. Lovastatin (high cholesterol)
49. Crestor® (high cholesterol)
50. Trimethoprim (bacterial infection)

AARP. The 50 Most Prescribed Drugs. 2009 Oct. http://pubs.aarp.org/aarpbulletin/200910?pg=55#pg55.

Please see registered trademark ownership in Appendix 1.

Thyroid Management Today

The thyroid disorder population is grossly underserved in both the conventional and alternative healthcare systems in the United States today. Practitioners within the conventional healthcare system simply measure TSH and provide synthetic hormones if it is abnormal. If any other symptoms persist after a normalization of their TSH, each existing symptom is then medicated. The question of asking why the patient has become hypothyroid is rarely asked. Furthermore, the question of why the patient may have persistent thyroid symptoms after normalizing the TSH is also rarely asked. This leads to millions of thyroid sufferers who continue to suffer. This takes place partly for two reasons. First, the conventional healthcare system has established standard-of-care guidelines that are the expected policy of hospitals and insurance plans. Second, deviation from the standard-of-care guidelines decreases peer acceptance and increases liability for the hospital or practicing physician.

The alternative healthcare approach to thyroid disorders is also grossly underserving the thyroid population. Many alternative healthcare professionals have simplified thyroid management to such things as iodine deficiency, tyrosine deficiency, or heavy metal toxicity. Additionally, many alternative healthcare professionals do not conduct laboratory testing and use unreliable testing procedures, such as auxiliary temperature testing, iodine skin absorption, etc. Many alternative healthcare professionals cannot see a world that exists for supporting the thyroid beyond iodine, tyrosine, and thyroid glandulars. Nor do they objectively measure change.

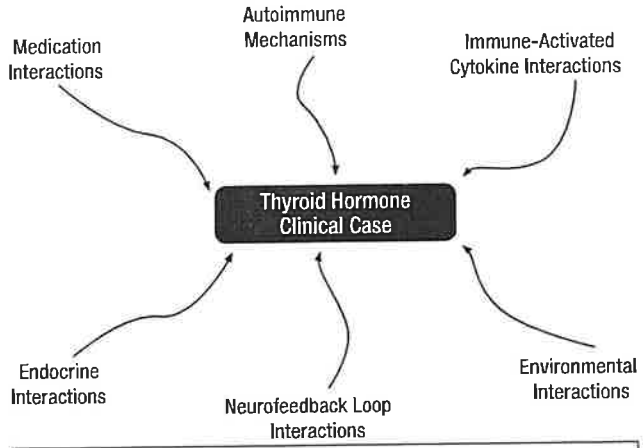
What Is the Definition of Functional Hypothyroidism?

Two completely different definitions of functional hypothyroidism exist in healthcare today. One is used by conventional medicine and the other is used by alternative medicine. In alternative medicine, functional hypothyroidism is typically defined as a thyroid condition that it not progressed enough to be considered a hypothyroid disease as identified by laboratory tests, but as a subtle disorder that can be supported nutritionally to improve thyroid function and health. Interestingly, the definition used by conventional medicine is almost the opposite. Conventional medicine defines functional hypothyroidism as a hypothyroid condition that does not present with any subjective thyroid symptoms.

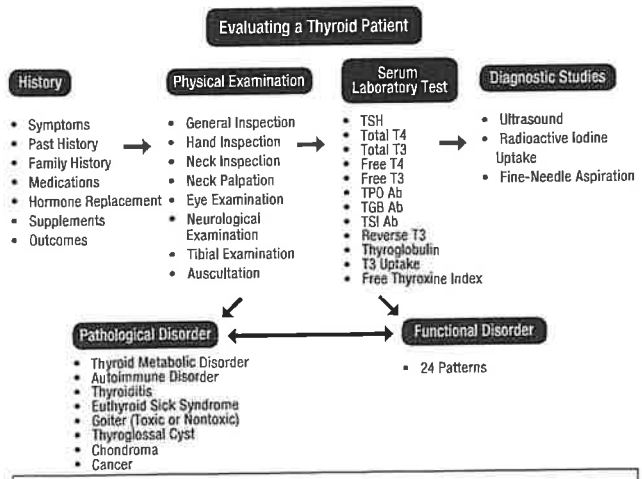
Which definition is correct? They are both correct depending on the application. The alternative medicine definition has merit because, even to this day, there

is no general consensus for thyroid ranges. Every major medical and endocrine group, nationally and internationally, has different opinions. However, in this course, we use very narrow ranges called functional ranges. These narrow ranges provide for the slightest changes that may be classified as functional hypothyroid by the alternative medicine definition.

The conventional medical definition of hypothyroidism is very common. Many patients are clearly hypothyroid, even when evaluated with very broad reference ranges, and do not present with any thyroid symptoms. They have dysfunction in their thyroid gland but have good compensatory physiological mechanisms that may lead to limited thyroid symptoms. However, the point is that they do have something wrong with their thyroid. These types of functionally hypothyroid patients are typically missed, unless routine thyroid laboratory tests are conducted during an annual physical or general screening blood chemistry.



For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.



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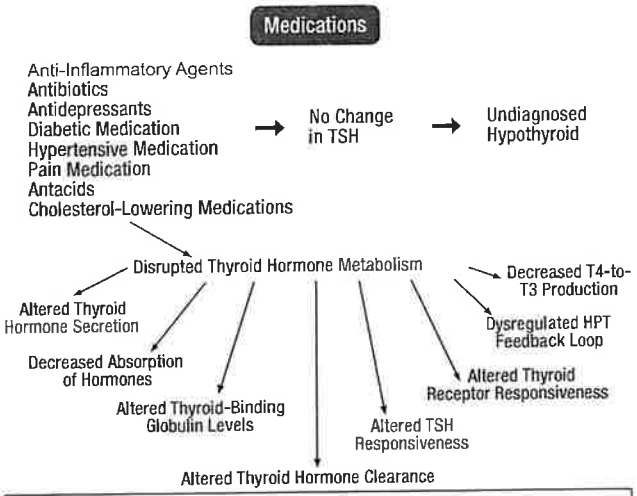

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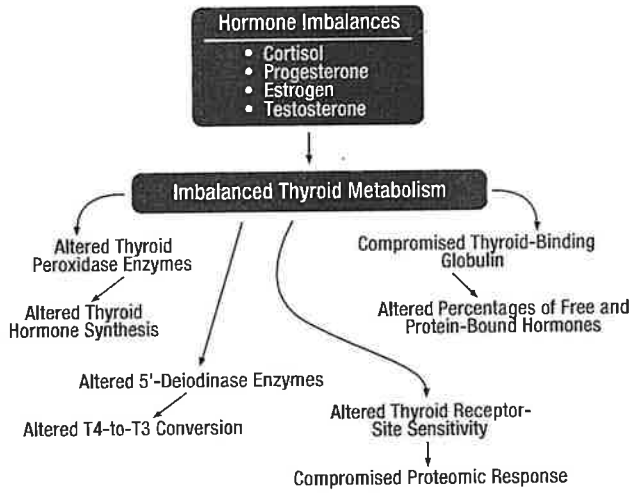
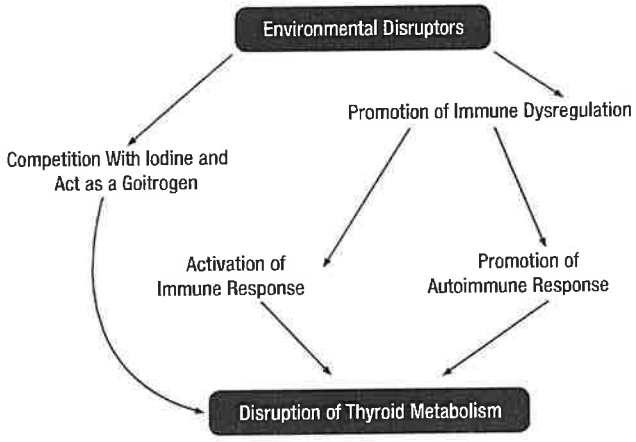
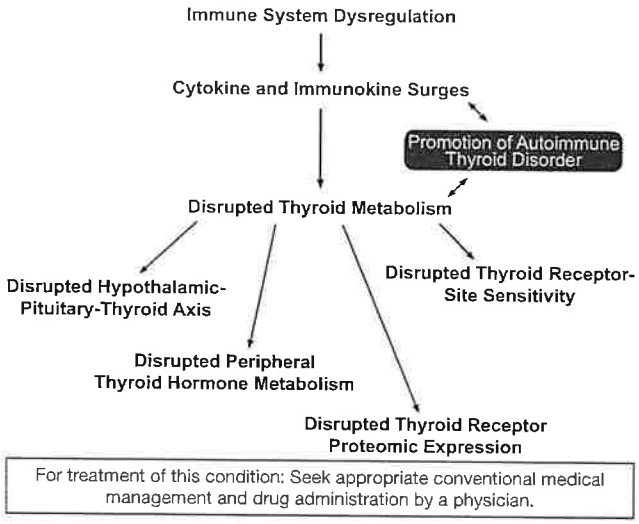
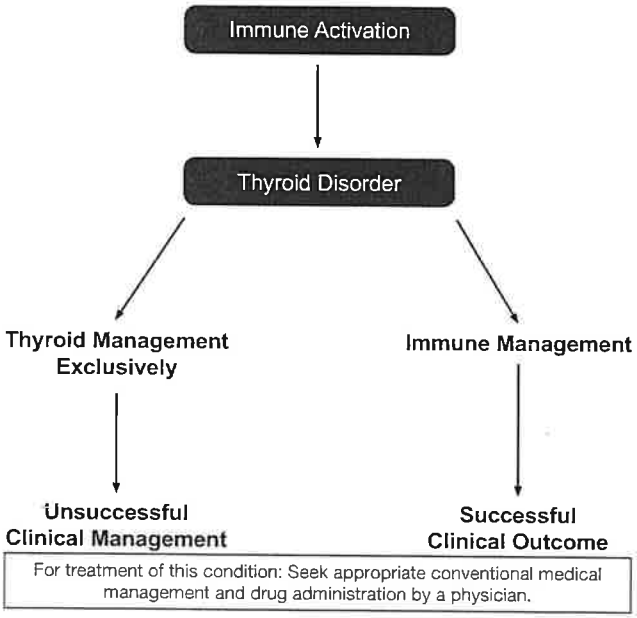








Where a condition is identified or a lab result is outside the normal range: Patient should seek appropriate conventional medical management and drug administration by a physician.

Case Example

Case 1

History

A 37-year-old female assistant financial advisor came into the office with complaints of deteriorating mental health (mental clarity, poor memory, etc), infertility, Hashimoto's hypothyroidism, high blood pressure, water retention, and eczema.

Her symptoms started when she was a graduate student in the UCSD PhD program in 1995. During her third year, she could no longer focus or learn. She had to drop out of the PhD program due to mental deterioration. During this time, she was having episodes of swelling, fatigue, and severe eczema. The symptoms continued until 2003, when she started to develop panic attacks and became intolerant to caffeine. She found that even small amounts of caffeine, such as in decaffeinated coffee or chocolate, caused her symptoms to flare up, making her miserable for three days, on average. In 2008, after several episodes of faulty ovulation, a TSH test was finally conducted and showed a TSH of 73. She was prescribed levothyroxine. Almost immediately after starting the levothyroxine, she had alternating days of hyperactivity and depressed apathy. She couldn't think straight in either condition. After three weeks, she was absolutely miserable.

She finally pursued care outside of her HMO and worked with a progressive MD who placed her on Armour®. She felt much better after switching to Armour®. Her new doctor monitored her temperature and decided that since it was low, she needed extra hormones, so he also placed her on Cytomel®. He also placed her on hydrocortisone for adrenal support. She had not lost any weight since starting any of the thyroid hormones. She did not eat large quantities of food and exercised 3-5 times per week. Her memory was terrible. At this point, she started fertility treatments. She was given 2-4 mg of synthetic estrogen and 200 mg of a progestin suppository in a fertility treatment cycle, as well as varying doses of Clomid® and Repronex®, with HCG stimulation shots.

Out of desperation, she started to educate herself. She came across a book called, *Why Do I Still Have Thyroid Symptoms When My Lab Tests Are Normal?* Based on recommendations in the book, she had her doctor run her TPO antibodies. Her TPO antibody test was significantly elevated at 1,736. She became a patient in our office in October of 2009.

Questions

- What key findings are important from her history?
- What does her intolerance to caffeine suggest?
- What is her mechanism for mental deterioration?
- What is her mechanism for infertility?
- What is her mechanism for eczema?
- What is the effect of hydrocortisone on her metabolism?
- What are the effects of the fertility treatment on her metabolism?
- What do her responses to the hormone replacement therapy tell you?

Examination

Evaluation of her forms indicated she had very poor handwriting. When questioned about her handwriting, she stated that it had been getting worse every year, but especially over the past three years. Visual inspection of her face and posture demonstrated severe facial paresis on the right side, right ptosis, and a wide-stance gait when walking. She had difficulty repeating numbers when asked to remember a set of them. She had elevated blood pressure and bilateral resting and intention tremors. Heart auscultation revealed tachycardia. Carotid artery auscultation revealed a bruit over the right carotid artery. She had significant difficulty with Romberg's and balance testing bilaterally.

Questions

- What do her poor handwriting and the rate of her progression indicate?
- What does her facial tone indicate?
- What do her wide-stance gait and balance problems indicate?
- What does her difficulty repeating numbers back suggest?
- Why does she have tachycardia?
- Why does she have bilateral resting and intention tremors?
- What does the right carotid artery bruit indicate?
- What would be your next step in this case?

Thyroid Physiology and Biochemistry

It would be impossible to truly understand and appreciate thyroid disorders, thyroid symptoms, thyroid signs, thyroid metabolic patterns, and thyroid laboratory findings without a thorough understanding of basic thyroid physiology and biochemistry. Clinicians today typically develop a strong grasp of physiology and biochemistry while they receive their training and prepare for licensing and board examinations, and then have the information slowly deteriorate from their recall after each year of practice. Both traditional and alternative medicine is based on established protocols and standards of care. These practices may help serve the basic patient, but they are not adequate for the complex cases that require the healthcare professional to think outside the box.

In this section, we review the hypothalamic-pituitary-thyroid axis, thyroid hormone synthesis, peripheral hormone conversions, thyroid hormone binding and unbinding, receptor-site responses, and basic concepts of the basal metabolic rate. Before we continue, please take a look at the 10 signs for proper thyroid hormone synthesis and ask yourself if you would be able to identify defects in any of these. Do they require the practitioner to think beyond simple models? The goal of this course is to look critically at as many known mechanisms, not just for established thyroid conditions with ICD.9 codes, but the physiology of healthy thyroid hormone synthesis and receptor-site expressions.

10 Metabolic Signs for Proper Thyroid Hormone Synthesis and Activity

- Proper serotonergic and dopaminergic activity in the hypothalamus
- Proper pituitary responses
- Proper thyroid-peroxidase enzyme activity
- Proper iodine uptake to the thyroid
- Proper amino acid synthesis of tyrosine
- Proper attachment and detachment of thyroid-binding proteins
- Proper 5'-deiodinase enzyme activity
- Proper intestinal sulfatase and acetylation enzyme activity
- Proper nuclear receptor-site sensitivity responses
- Proper thyroid proteomic responses

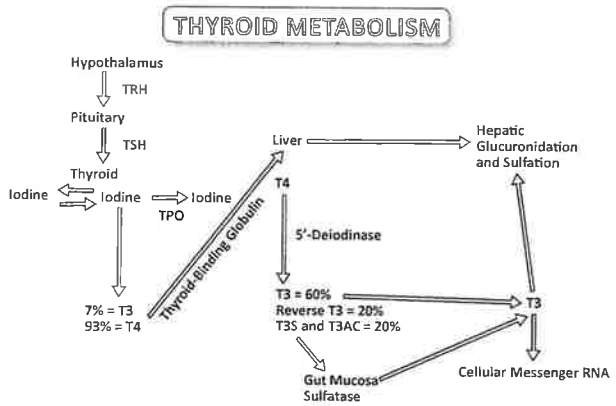
Thyroid Central and Peripheral Physiology

The thyroid axis begins with the hypothalamus. Within the hypothalamus, a group of nuclei called the paraventricular nucleus (PVN) regulates endocrine output, such as thyrotropin-releasing hormone (TRH). These hypothalamic nuclei are primarily modulated by serotonergic and dopaminergic signaling pathways. Therefore, chemical factors that disrupt these neurotransmitter substances may have an influence on the thyroid axis. Once the PVN has been excited by neurotransmission pathways, the hypothalamus releases TRH to stimulate the pituitary.

The pituitary, upon stimulation from TRH, releases thyrotropin, also know as thyroid-stimulating hormone (TSH). Thyroid-stimulating hormone will then stimulate metabolic activity of the cells and also stimulate the enzyme thyroid peroxidase (TPO) to produce thyroxine (T4) and triiodothyronine (T3) by transporting iodine into the thyroid. Thyroid peroxidase is involved in the formation of T4 and T3 as it catalyzes the oxidation of iodine using hydrogen peroxide. The thyroid gland produces 94% of the available T4 and 7% of the available T3; T4 is inactive, and T3 is the active form of thyroid hormone. Therefore, the majority of hormone production at the thyroid is inactive T4. Additionally, the hormones produced by the thyroid gland are bound to a protein (thyroid-binding globulin, albumin, and transthyretin).

Once the thyroid has produced T4, it is metabolized peripherally into a combination of T3 hormones by the enzyme 5'-deiodinase, mostly at the liver. Under normal circumstances, about 40% of the available T4 is converted into T3, 20% is converted into reverse T3, which is irreversibly inactive, and 20% is converted into T3 sulfate (T3S) and triiodothyroacetic acid (T3AC). Triiodothyronine sulfate and T3AC are inactive thyroid hormones until they circulate into the gastrointestinal tract and are acted upon by intestinal sulfatase to become active T3. Gastrointestinal sulfatase activity is dependent upon healthy gut microflora. At some point, active, unbound (or free) T3 binds to the nuclear receptor of the target tissue and promotes mitochondrial activity and the activation of the cellular basal metabolic rate.

In summary, thyroid hormone physiology starts with brain synaptic activity, goes to the pituitary, then to the thyroid gland, then to peripheral tissues, such as the liver and gastrointestinal tract, before it has a chance to become activated and bind to receptor sites. Several mechanisms are important for ideal thyroid hormone synthesis and activation. These include proper serotonergic and dopaminergic activity, proper pituitary responses, proper TPO enzyme activity, proper iodine uptake to the thyroid, proper amino acid synthesis of tyrosine, proper attachment and detachment of thyroid-binding proteins, proper 5'-deiodinase enzyme activity, proper intestinal sulfatase and acetylation enzyme activity, proper nuclear receptor-site sensitivity responses, and proper thyroid proteomic responses. Any defect along this chain can decrease the potential for thyroid hormone synthesis and effect on bodily tissues.



Thyroid Hormone Binding and Free Hormones

Thyroid hormones can be either "protein-bound" or "free." As these names imply, a protein-bound hormone is a thyroid hormone (T4 or T3) bound to a protein. These proteins include thyroxine-binding globulin (TBG), transthyretin (TTR), thyroxine-binding prealbumin (TBPA), and para-albumin. Hormones that are not bound to any protein are termed free-fraction hormones, such as free T3 or free T4. Only 0.03% of the total T4 in the body is free, and only 0.3% of the total T3 in the body is free. Almost 99% of the thyroid hormones in the body are protein-bound.

Percentage Values of Bound & Unbound Thyroid Hormones

Type	Percent
Bound to thyroid-binding globulin (TBG)	70%
Bound to transthyretin (TTR) or "thyroxine-binding prealbumin" (TBPA)	10%-15%
Para-albumin	15%-20%
Unbound T4 (free T4)	0.03%
Unbound T3 (free T3)	0.3%

Only free-fraction hormones can bind to receptor sites and have an impact on metabolism. Protein-bound hormones are structurally too large to attach to a receptor site and therefore have no influence on metabolism until they lose their protein and become a free hormone. Hormones have proteins attached to them and are unable to bind to receptor sites so they can be transported to peripheral tissues throughout the body. When the thyroid gland synthesizes T4 and T3, they are all protein bound. At some point, under mechanisms that are not clearly understood, the hormone loses its protein and becomes free. Enzymes in peripheral tissues, the liver and the kidneys unbind thyroid hormones into a free state. Once thyroid hormones are free, they circulate in the blood stream for 15 minutes to several hours until they reach a tissue where they bind to the thyroid receptor sites and have an effect.

Understanding these mechanisms is very important when evaluating thyroid markers. For example, protein-bound T4 and T3 are more accurate markers for the quantity of thyroid hormones the thyroid gland is producing. Free T4 and free T3 are more accurate markers for the quantity of thyroid hormones that have the ability to bind to receptor sites. A discrepancy between the quantity of protein-bound to free-fraction hormones surely indicates an influence on the thyroid-binding occurring.

When evaluating thyroid hormone laboratory results, you must see the prefix "free" before T4 and T3 to assume it is a free fraction hormone, such as free T4 or free T3. If you do not see the prefix "free," such as serum T4 or serum T3, you are evaluating protein-bound hormones. If the laboratory lists the thyroid hormones as "total," such as total T3 or total T4, they are referring to both the free- and protein-bound. However since 99.3% of total hormones are protein-bound, you must clinically

consider "total" as a protein-bound marker. Changes in quantity of the 0.7% of the hormones will not skew the total T4 or total T3 range.

Basic Functions of Thyroid Hormones

Thyroid hormones "act on the body to increase the basal metabolic rate, affect protein synthesis, and increase the body's sensitivity to catecholamines. They are essential to proper development and differentiation of all cells of the human body. Thyroid hormones regulate protein, fat, and carbohydrate metabolism, affecting how cells use energetic compounds for ATP production." All tissues of the body (brain, heart, bone, muscle, etc) have thyroid receptor sites, and since thyroid hormones are essential in cellular bioenergetics for ATP production, dysregulation of thyroid hormone (in excess or deficiency) leads to numerous areas of dysfunction.

Basal Metabolic Rate

"Basal metabolic rate (BMR) is the amount of energy expended while at rest in a neutrally temperate environment, in the postabsorptive state" (meaning the digestive system is inactive, which requires about 12 hours of fasting in humans). "The release of energy in this state is sufficient only for the functioning of the vital organs." Thyroid hormones are the most impactful hormones on the body's BMR. In states of hypofunction and decreased metabolic rate, there will be decreased muscle activity leading to poor muscle endurance, slower heart rate, greater bone loss, decreased gastrointestinal motility, etc. The opposite would occur with thyroid hyperfunction and increased BMR.

Thyroid Hormone Production

"Thyroxine (3,5,3',5'-tetraiodothyronine) is produced by follicular cells of the thyroid gland. It is produced as the precursor thyroglobulin (this is *not* the same as TBG), which is cleaved by enzymes to produce active T4. Thyroxine is produced by attaching iodine atoms to the ring structures of tyrosine molecules. Thyroxine (T4) contains four iodine atoms. Triiodothyronine (T3) is identical to T4, but it has one less iodine atom per molecule."

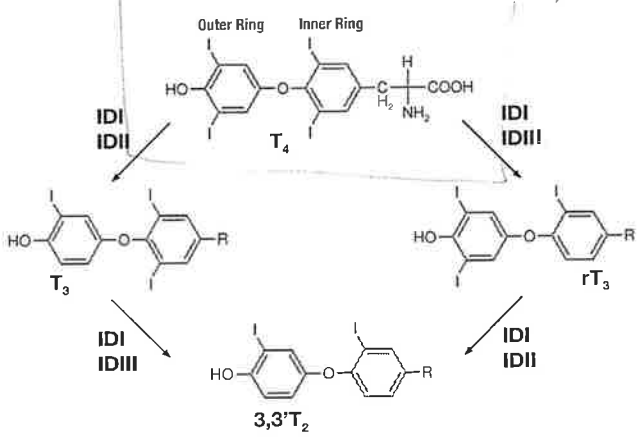
"Iodide is actively absorbed from the bloodstream by a process called 'iodine trapping'" and concentrated in the thyroid follicles. "Via a reaction with the enzyme thyroperoxidase, iodine is covalently bound to tyrosine residues in the thyroglobulin molecules, forming monoiodotyrosine (MIT) and diiodotyrosine (DIT). Linking two moieties of DIT produces thyroxine. Combining one particle of MIT and one particle of DIT produces triiodothyronine.

DIT + MIT → reverse T3 (biologically inactive)

MIT + DIT → triiodothyronine (usually referred to as T3)

DIT + DIT → thyroxine (referred to as T4)

There are three types of enzymes that can deiodinate thyroxine. Iodothyronine deiodinase type I (IDI) is commonly found in the liver, kidney, muscle tissue and thyroid gland, and type II (IDII) in the testis and thyroid. Type III (IDIII) is found in the fetal tissue, brain matter, and the placenta. Type II can only deiodinate the outer ring of the prohormone thyroxine or the metabolically inactive reverse triiodothyronine. Type III can only deiodinate the inner ring of thyroxine or triiodothyronine.Type I can deiodinate both rings."



T3 is a metabolically active hormone that is produced from T4. Thyroxine is deiodinated by two deiodinases to produce the active T3:

1. *Type I is present within the liver and accounts for 80% of the deiodination of T4.*
2. *Type II is present within the pituitary*

T4 is synthesized in the thyroid gland follicular cells as follows:

1. "The Na+/I- symporter transports two sodium ions across the basement membrane of the follicular cells along with an iodine ion. This is the secondary

active transporter that utilizes the concentration gradient of Na+ to move I- against its concentration gradient."

2. "I- is moved across the apical membrane into the colloid of the follicle."
3. "Thyroperoxidase oxidizes two I- to form I2. Iodide is non-reactive and only the more reactive iodine is required for the next step."
4. "The thyroperoxidase iodinates the tyrosyl residues of the thyroglobulin within the colloid. The thyroglobulin was synthesized in the ER of the follicular cell and secreted into the colloid."
5. "Thyroid stimulating hormone released from the pituitary gland binds the TSH receptor (a Gs protein-coupled receptor) on the basolateral membrane of the cell and stimulates the endocytosis of the colloid."
6. "The endocytosed vesicles fuse with the lysosomes of the follicular cell. The lysosomal enzymes cleave the T4 from the iodinated thyroglobulin."

"These vesicles are then exocytosed releasing the thyroid hormones." In the follicular lumen, tyrosine residues become iodinated. This reaction requires hydrogen peroxide. Iodine bonds carbon 3 or carbon 5 of tyrosine residues of thyroglobulin in a process called organification of iodine. The iodination of specific tyrosines yields MIT and DIT. One MIT and one DIT are enzymatically coupled to form T3. The enzyme is TPO.

Thyroid Hormone Receptors

Thyroid receptors are nuclear receptors and function as hormone-activated transcription factors and thereby act by modulating gene expression. "In contrast to steroid receptors, thyroid receptors bind to DNA in the absence of hormones, usually leading to transcriptional repression. Hormone binding is associated with conformational change in the receptor that causes it to function as a transcriptional activator."

Currently, there are four identified thyroid receptors: alpha-1, alpha-2, beta-1, beta-2. The different forms of thyroid receptors have patterns of expression that vary by tissue and by developmental stages. "The presence of multiple forms of thyroid hormone receptors, with tissue- and stage-dependent differences in their expression, suggests an extraordinary level of complexity in the physiological effects of thyroid hormone."

Thyroid Receptors and Tissues

- **TR-alpha-1—widely expressed and especially high expression in cardiac and skeletal muscles**
- **TR-alpha-2—homologous with viral oncogene c-erb-A, also widely expressed but unable to bind hormones**
- **TR-beta-1—predominately expressed in brain, liver, and kidney**
- **TR-beta-2—expression primarily limited to the hypothalamus and pituitary**

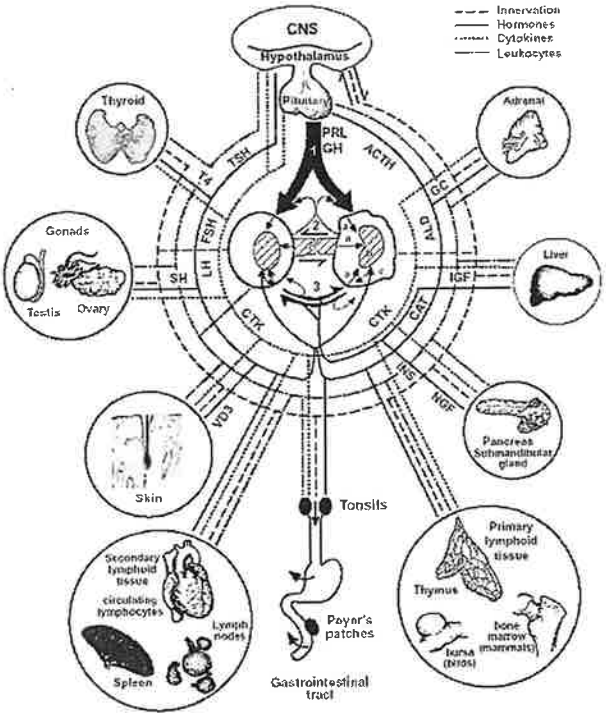
"A number of humans with a syndrome of thyroid hormone resistance have been identified and found to have mutations in the receptor beta gene that abolish ligand binding. Clinically, such individuals show a type of hypothyroidism characterized by goiter, elevated serum concentrations of T3 and T4, and normal levels of TSH. Typically, more than half of the children with thyroid receptor-site mutations develop learning disorders. Mice with targeted deletions in thyroid receptor genes have provided additional understanding of the possible roles of different forms of TRs. Knockout mice that are unable to produce the alpha-1 receptor showed subnormal body temperature and mild abnormalities in cardiac function. Other mice, which lacked expression of both alpha isoforms, were severely hypothyroid and died within the first few weeks of life. Mice with disruptions of the entire beta2 expression had elevated TSH, but normal hearing. Such experiments enable scientists to determine which functions of the different receptors are redundant and which are not."[4-6]

Thyroid-Immune Crosstalk

The evidence clearly demonstrates that there is functional crosstalk between the immune system and the thyroid. Thyroid hormones are not only used to support immune cells' basic metabolic rate, but they also exhibit influences on cellular messenger systems that promote modulation of T- and B-cell activity, cellular apoptosis, cytokine release, and other immunokine responses. Therefore, thyroid hormones are classified as immune system modulators, and thyroid function and metabolism should be considered in cases of immune dysregulation or immune challenges.

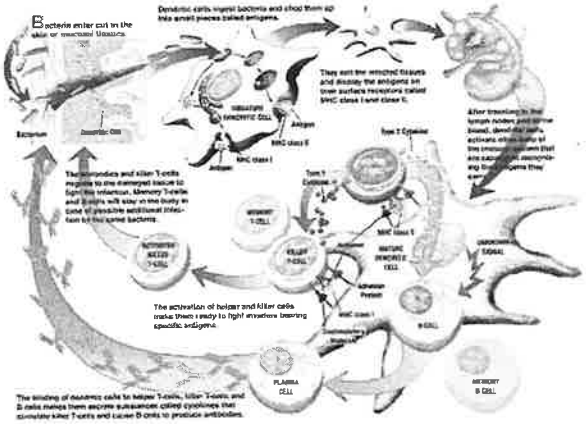
On the other hand, immune-signaling mediators such as cytokines and adjuvants have been found to disrupt all aspects of the hypothalamic-pituitary-thyroid (HPT) axis, peripheral metabolism, and thyroid receptor-site transcriptional expressions. Immune disorders have great potential to disrupt thyroid physiology and promote concomitant dysfunction in cellular metabolism. It is important for the practicing clinician to understand the thyroid-immune and immune-thyroid interactions. Part I of this section on immunology and thyroid interactions will describe the mechanisms of how the immune system can influence thyroid metabolism from the HPT axis all the way to the receptor site. Part II will review the influence of thyroid hormones and various aspects of T-cell, B-cell, dendritic cell, natural killer cell, and other immunological activities. Part III of this section on immunology and thyroid will summarize this thyroid-immune crosstalk.

The Neuroimmune Regulatory Network



Source: Berczi, I. *NeuroImmune Biology*. Vol 3. Waltham, MA: Elsevier; 2003:584-585.

Interactions of cellular and humoral immunity as defense against invaders



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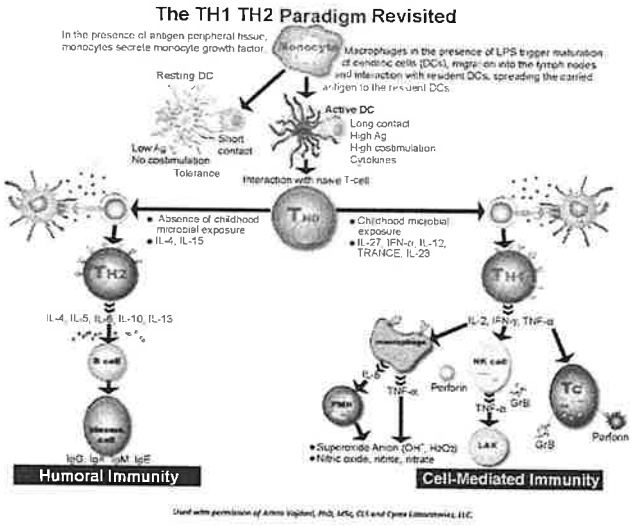


Part I: Immune Activity on the Thyroid

The immune system has profound impacts on the endocrine system, especially in regard to the thyroid, from both autoimmune and non-autoimmune perspectives. Certain patterns of thyroid dysfunction strongly suggest an evaluation of the immune system. The immune system appears to influence the hypothalamus-pituitary-thyroid axis in relation to TSH release and T3 levels from a nonautoimmune perspective. From an autoimmune perspective, the most common cause of hypothyroidism in the United States today is hypothyroidism secondary to Hashimoto's. Therefore, anytime an individual has an elevated TSH or has been diagnosed with hypothyroidism, antibodies against the thyroid (TPO Ab and TGB Ab) should be considered. Hashimoto's is generally a TH-1– predominate attack against self-thyroid tissue, and once the genes for this disease turn on, the individual becomes increasingly susceptible to cytokine fluctuations for further immune attack on self-thyroid tissue.

In this section of the notes, the focus is on the immune system's role on the thyroid with respect to a nonautoimmune thyroid response. A detailed review of the autoimmune thyroid pathophysiology will be discussed in other sections of the manual. Immune-related mechanisms that dysregulate thyroid function are common in the chronically ill, chronically inflamed, chronically antigen-challenged (hepatitis, Lyme, etc), and in systemic autoimmune responses. Immune adjuvants such as lipopolysaccharides related to gut microflora integrity and thyroid metabolism will be reviewed. However, we will first start with a basic review of cytokines.

Cytokines are messenger proteins that are released by immune cells found in different tissues in the body. Activation of cytokines, despite the tissue of origin, has the potential to produce effects on metabolism and physiology throughout the body. Cytokine molecules have effector influences upon a diverse network of neurological, endocrinological, and immunological responses. Cytokines are classified into two distinct categories labeled as T-helper 1 (TH-1) or T-helper 2 (TH-2). They are also known as pro-inflammatory (TH-1) T-cell–mediated or anti-inflammatory (TH-2) B-cell–mediated compounds. TH-1 cytokines include interferon (IFN) gamma, interleukin-2 (IL-2), and tumor necrosis factor (TNF) alpha. TH-2 cytokines include IL-4, IL-5, IL-6, IL-10, and IL-13. There are many cytokines that have pluripotentiality, meaning that they may exhibit both T-cell–and B-cell–mediated responses; however, most of the efforts conducted in the scientific community compare TH-1 and TH-2 subsets to autoimmune and physiological disorders. Further research has also identified TH-3 and TH-17. TH-3 subsets are related to regulatory T-cell function (CD25+4). Regulatory T-cells are used to modulate the balance of TH-1 and TH-2 responses. TH-3 activity is promoted by TGF-beta. TH-17 is T-cell activity related to the immunological destructiveness potential of TH-1 and TH-2 shifts. The greater the TH-17 activity, the greater potential for immunological destruction is present. TH-17 activity is promoted primarily by IL-6 and TH-17 activity is measured by IL-17. There is an independent loop between TH-1, TH-2, TH-3, and TH-17 cytokine responses that will be elaborated on more in the autoimmune section of this manual. It is best to start with a basic understanding of cytokines unrelated to autoimmune responses, but rather related to immunological activity.

Cytokines promoted by immunological activation have demonstrated profound and specific influences on thyroid metabolism and physiology that would be appropriately classified under the title of euthyroid sick syndrome or nonthyroidal illness (NTI). Nonthyroidal illness refers to thyroid conditions in which the thyroid gland itself is not diseased, but rather chemical imbalances are promoting a loss of healthy thyroid hormone metabolism. It appears that cytokines have

potentials to disrupt the hypothalamus-pituitary-thyroid axis and down-regulate the expression of T4 into active T3 peripherally by type 1 and type 2, 5'-deiodinase enzymes.

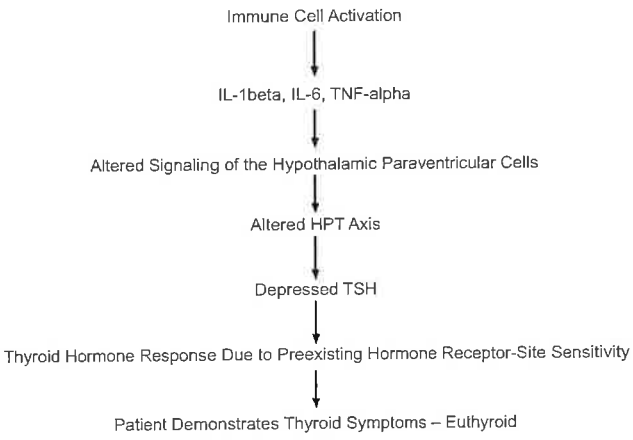
Cytokines and the Hypothalamic-Pituitary-Thyroid Axis

Cytokines IL-1 beta, IL-6, and TNF-alpha have been shown to down-regulate the hypothalamic-pituitary-thyroid axis. Administrations of these cytokines into either the peripheral subcutaneous tissue, the peritoneal cavity, or the third ventricles of animals have produced reductions of TSH.[7-12] These reductions in TSH are only evident when evaluated in comparison to pre-existing TSH results and control groups. Tumor necrosis factor–alpha, a peptide secreted from stimulated macrophages during inflammatory conditions, has the ability to impair the hypothalamic-pituitary-thyroid axis. An animal study was conducted in which subjects were injected with TNF-alpha to be followed be thyroid biochemistry assay. One single injection of TNF-alpha reduced serum TSH, T3, free T4, T3, and hypothalamic TRH compared to the control for five days. TNF-alpha was found to inhibit hepatic 5'-deiodinase, reduce thyroid hormone uptake, reduce thyroidal release of T4 and T3 in response to TSH, alter the glycosylation of TSH, and suppress hypothalamic TRH release.[13]

The TSH range used in the standard healthcare system is so extremely broad that these patterns are typically not considered during clinical work-up. Since the standard TSH range is very wide, individuals who have issues resulting from cytokine suppression of TSH will not be assessed properly. These individuals will suffer from thyroid symptoms and also experience receptor-site resistance, despite a TSH that is within the wide reference range.

Practitioners who evaluate blood chemistry functionally are always considerate of pituitary suppression by using a TSH functional range of 1.8 as the low end of the range, compared to the typical laboratory range of 0.5. The TSH low-end range does not reflect ideal physiology. It is skewed by the bell-curve analysis that is used to formulate the range, since many subjects in the bell curve analysis are taking thyroid hormones that have suppressive influences on TSH. In functional evaluations of TSH, if it is below 1.8 and the patient has thyroid symptoms, the possibility of this pattern is suggested. Evaluation of cytokines (IL-1, TNF, IFN, IL-6) associated with euthyroid suppression of ideal hypothalamic-pituitary-thyroid axis feedback loop coordination should be considered.

Cytokines and the HPT Axis

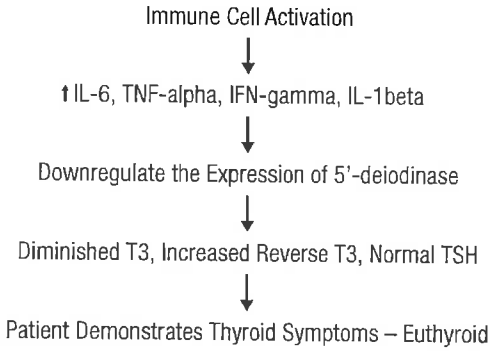


Cytokines and Peripheral Thyroid Hormone Conversion

Cytokines may not only have the potential to impact the hypothalamic-pituitary-thyroid axis, but also the ability to disrupt peripheral conversion of inactive T4 into active T3. The enzyme 5'-deiodinase catalyzes the conversion of T4 into T3 in peripheral tissues away from the thyroid. Both TH-1 and TH-2 cytokines—IL-6, TNF-alpha, INF gamma, and IL-1 beta—have demonstrated the ability to down-regulate thyroid conversion potentials of T3.[14-18] Patients with NTI have demonstrated an inverse relationship between serum T3 and IL-6 levels.[19] The administration of cytokines to healthy human subjects has shown to lead to immediate changes in thyroid hormone levels. Exposure to IL-6, TNF-alpha, and IFN-alpha to healthy subjects resulted in a rapid reduction of serum T3 and TSH levels and in increase in reverse T3 concentrations, while T4 and free T4 levels were minimally altered.[20-22]

Clinically, it is important to note that T3 levels have very little influence on the TSH negative feedback loop. Therefore, when T3 levels decline from a cytokine 5'-deiodinase mechanism, the TSH will typically not be elevated. If TSH is used as an isolated marker to screen for thyroid disorders, the T3 under-conversion pattern promoted by an elevation of cytokines may be overlooked. It should be standard to measure T3 when evaluating the thyroid. A pattern of normal TSH, normal T4, and a depressed T3 strongly suggests down-regulation of the 5'-deiodinase enzyme.

Cytokines and Peripheral Thyroid Hormone Conversion



Lipopolysaccharides and Thyroid Metabolism

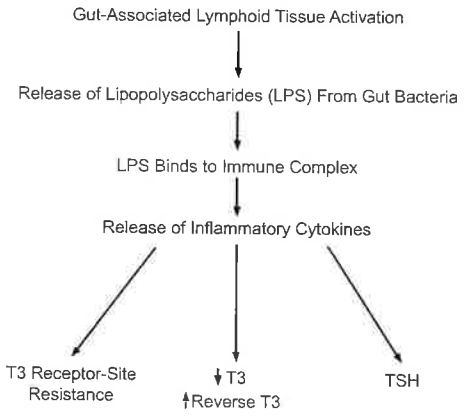
"Gastrointestinal microorganisms such as lipopolysaccharides (LPSs) are adjuvants that impact thyroid physiology" at the level of the central and peripheral systems. The LPSs released from gram-negative bacteria are considered endotoxins that may be used as adjuvants in the immune response. Adjuvants are agents that cannot potentiate an immune response alone, but with the orchestration of other cellular agents can influence physiological responses. Lipopolysaccharides acts as the prototypical endotoxin because it binds the CD14/TLR4/MD2 receptor complex, which promotes the secretion of pro-inflammatory cytokines in many cell types, but especially in macrophages.[23]

LPS has potential to influence the thyroid at all levels of function, including reducing thyroid hormone levels, diminishing the expression of the TR, increasing inactive reverse T3 levels, decreasing TSH concentrations, and promoting autoimmune thyroid disorders. A study of [24] healthy subjects who underwent IV injection of an LPS endotoxin demonstrated diminished levels of serum T4, serum free T3, and serum TSH, and inactive reverse T3 was elevated when compared with controls.[24] A similar study was conducted with 18 healthy subjects who underwent intravenous injections with an LPS endotoxin from Escherichia coli (EC-5). The LPS-challenged subjects demonstrated decreased levels of serum T4, serum free T4, T3, TSH, and a rise in inactive reverse T3.[25] These studies suggest that an immune challenge and abnormal gastrointestinal microorganism environments may promote abnormal central pituitary release of TSH as well as potential peripheral thyroid hormone production. Lipopolysaccharides have been shown to contribute to the pathogenesis of nonthyroidal illness of the central part of the hypothalamus-pituitary-thyroid axis. Lipopolysaccharide administration to study subjects decreased serum thyroid hormone levels and diminished liver deiodinase (D1) mRNA expression. Lipopolysaccharide administration was also found to disrupt pituitary-thyroid coordination by decreasing the expression of D2 mRNA expression.[26]

Lastly, the promotion of elevated LPS activity as a result of infection demonstrated a reduction of TR expression. In gel shift experiments, retinoid X receptor (RXR)/TR DNA binding was reduced by 60 and 77% respectfully, 4 and 16 hours after LPS administration.[27] These dramatic changes occurring at the receptor sites cannot be recognized with routine thyroid testing and may explain euthyroid presentations clinically.

In conclusion, LPSs in combination with other variables have the potential to disrupt thyroid-hypothalamus-pituitary coordination, thyroid hormone to T3 conversion, the amount of thyroid hormones, and the expressiveness of thyroid receptors to thyroid hormones. The connections with euthyroid clinical presentation inflammation, gastrointestinal disorders, and immune activity are beginning to be understood better. Clinically, the immune system and its role in LPS activity cannot be overlooked with patient populations who present with thyroid symptoms and euthyroid laboratory presentations.

Lipopolysaccharides and Thyroid Metabolism



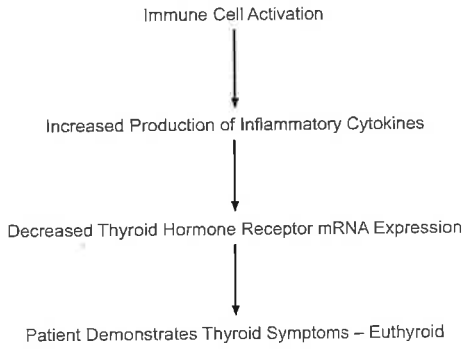
Cytokines and Thyroid Receptor-Site Expression

Inflammatory cytokines alter thyroid function and thyroid receptor-site proteomic responses.[28-30] Proinflammatory cytokines have been associated with nonthyroid illness in view of their capability to decrease D1 and TR-beta-1 mRNA expression. Proinflammatory cytokine induction leads to activation of the inflammatory pathways

nuclear-factor-kappa-B (NF-kB) and activator protein-1. "The proinflammatory cytokine interleukin1-beta, decreases thyroid hormone receptors beta1 mRNA in an NF-kB-dependent way, leading to decreased post-translational receptor activity."[31]

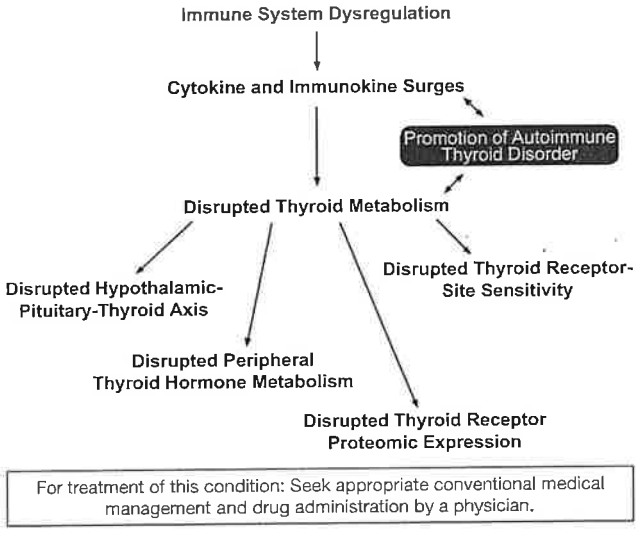
Clinically, the amount of receptor-site expression cannot be measured in private practice, and assumptions about thyroid receptor-site activity must be made with elevated cytokines. Active immune-related patterns may hinder thyroid metabolism and a careful evaluation of immune-related disorders must be provided to all individuals who present with thyroid symptomatology. A perfect example of this clinically is the autoimmune thyroid patient or a patient with an inflammatory disease who is taking replacement hormones but is still suffering from thyroid symptoms. In these scenarios, the elevations of the inflammatory immune cytokines are depressing thyroid receptor-site responsiveness, and the replacement hormones are creating a normal appearance of thyroid laboratory markers (TSH, T4, and T3). The concept of cytokines suppressing thyroid receptor-site responsiveness is not clinically acknowledged, and the patient ends up without proper management.

Cytokine and Thyroid Receptor-Site Expression



Summary of Immune Activity on the Thyroid

- Cytokine suppression of HPT coordination
- Cytokine suppression of TRH release
- Cytokine suppression of receptor glycosylation of TSH
- Cytokine suppression of thyroid hormone release
- Cytokine suppression of T4 conversion into T3
- Cytokine suppression of thyroid post-transcriptional receptor-site activity
- Cytokine suppression of thyroid hormone uptake
- LPS or adjuvant suppression of thyroid hormone secretion
- LPS or adjuvant promotion of decreased thyroid receptor expression
- LPS or adjuvant increased production of reverse T3 and decreased production of active T3
- LPS or adjuvant suppression of TSH production
- LPS or adjuvant promotion of autoimmune thyroid response LPS or adjuvant suppression of HPT coordination



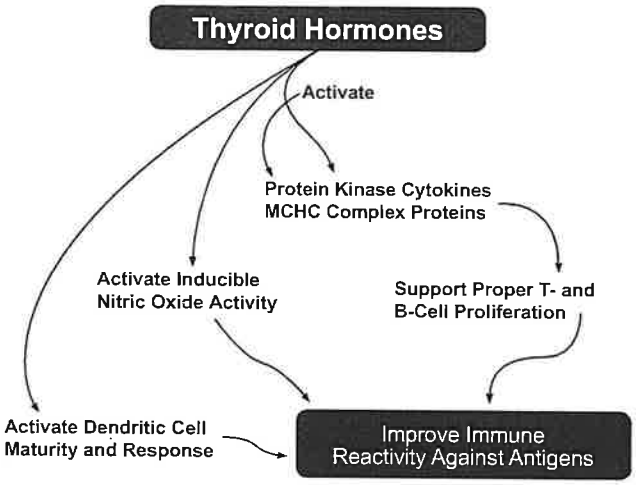
Part II: Thyroid Influence on Immunology

Immune Regulation and Thyroid Hormones

Thyroid hormones play critical roles in the differentiation, growth, and metabolism of immune system regulation. Researchers conducted an "integrative analysis of the role of the hypothalamic-pituitary-thyroid axis T- and B-lymphocyte activity. The participation of the protein kinase C (PKS) signaling pathway and the release of some cytokines upon antigenic stimulation were analyzed. Lymphocytes from hyperthyroid mice displayed higher T- and B-cell mitogen-induced proliferation, and those from hypothyroid mice displayed lower T- and B-cell mitogen-induced proliferation, compared with euthyroid animals. Reversion of hypothyroid state by T3 administration recovered the proliferative responses." These results demonstrate the importance of thyroid hormones in regulating lymphocyte reactivity.[32]

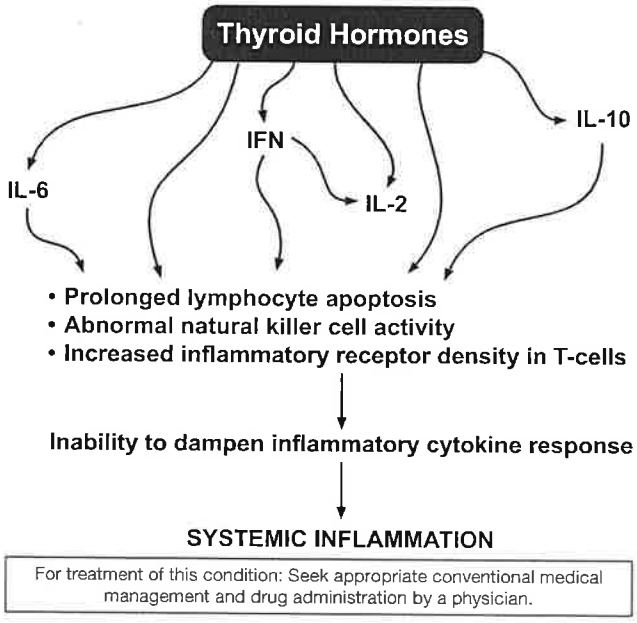
Thyroid hormones have also been shown to enhance T-helper–mediated stimulation of inducible nitric oxide

synthase and cell proliferation.[33] Evidence clearly demonstrates that there is functional crosstalk between immune and thyroid mechanisms. A study examined the presence of TRs and the impact of thyroid hormones in the physiology of dendritic cells, specialized antigen-presenting cells with the unique capacity to fully activate naïve T cells and orchestrate adaptive immunity. It was found that physiological levels of T3 stimulated the expression of dendritic cell maturation markers (major histocompatibility complex II, CD80, CD86, and CD40), markedly increased the section of interleukin-12, and stimulated the ability of dendritic cells to induce naive T-cell cultures. According to the study, "analysis of the mechanisms involved in these effects revealed the ability of T3 to influence the cytoplasmic-nuclear shuttling of nuclear factor-kappaB on primed dendritic cells." The researchers went on to say, "these results have profound implications in immunopathology, including cancer and autoimmune manifestations of the thyroid gland at the crossroads of the immune and endocrine systems."[34]



Inflammatory Modulation and Thyroid Hormones

Intimate relationships exist between thyroid hormones on immune cell modulation, and immune cell cytokine release on the thyroid axis. One study conducted on healthy individuals aged 55-70 years of age found "thyroid hormone concentration was positively associated with markers for inflammation, natural killer-like T (NKT) cells, expression of IL-6 by activated monocytes; expression of memory T-lymphocytes, memory T-helper lymphocytes, memory T-cytotoxic lymphocytes, and higher IL-2 receptor density on CD3+T-lymphocytes. Thyroid hormone concentration was inversely associated with early lymphocyte apoptosis and the ratio of naive-to-memory T-cytotoxic lymphocytes." The results of these findings conclude that thyroid hormones within normal physiological ranges are necessary to maintain lymphocyte subpopulations and to mediate inflammatory responses.[35] A different study looked at the production of T-lymphocytes in vivo and in vitro with thyroxine administration. The results found that thyroxine has the potential to dampen inflammatory cytokines such as IFN-gamma, IL-4, and IL-10.[36]

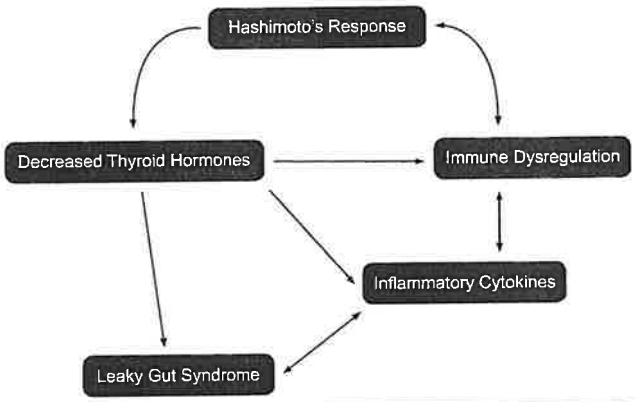


Intestinal Permeability and Thyroid Hormones

Thyroid hormones have powerful influences on the integrity of gastrointestinal mucosal tight junctions in both the stomach and small intestine. Lack of thyroid hormone stimulation to gastric and intestinal cells leads to ulcerations and intestinal permeability known as "leaky gut syndrome." Both thyroxine and triiodothyronine have been shown to protect the intestinal mucosal lining from stress-induced ulcer formation.[37] Additionally, endoscopic examination of gastric ulcers found low T3, low T4, and abnormal levels of reverse T3 on gastric ulcerative tissues.[38]

"Recent studies in athymic mice indicate that the neuroendocrine hormones TRH and TSH can significantly influence the development of lymphoid cells associated with intestinal intraepithelial lymphocytes (IEL)." Thyroxine has shown to prevent overexpression of IEL cells toward inflammation.[39] These findings implicate the connection between thyroid hormone activity and potential intestinal inflammation from the thyroid-gastrointestinal axis.

Animal studies with surgically induced hypothyroidism and then stressed with and without thyroid hormones have demonstrated that "low-circulating thyroid hormones have an ulcerogenic effect and that replacement of T3 in rats with surgical hypothyroidism reverses this effect."[40] It also appears that catecholamine levels in the gastric walls prevent ulcer formation. Catecholamines promote blood flow and repair in the gastric wall. In a 1976 study, "it was demonstrated in experiments on rats that induction of "stress" and "steroid" ulcer of the stomach was accompanied by reduction in adrenaline and noradrenaline content in the wall of the stomach. Induction of such ulcers in the animals with experimental disturbances of the thyroid gland function was also accompanied by a reduction in catecholamine content in the wall of the stomach."[41]



For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

TH-1/TH-2 Dysregulation and Hypothyroidism

An interesting study was conducted looking at the role of prophylactic L-thyroxine therapy in euthyroid Hashimoto's, since thyroid hormones have been shown to help modulate immune activity. Increased immune dysregulation leads to greater thyroid tissue attack and the progression of euthyroid Hashimoto's to hypothyroid Hashimoto's. The study evaluated 33 patients diagnosed with euthyroid Hashimoto's thyroiditis and randomized them into two groups. One group received prophylactic L-thyroxine treatment and the other was followed-up without treatment. Initial thyroid function tests, autoantibodies, ultrasonography, FNA biopsy, and peripheral blood lymphocyte subsets were similar in the two study groups. After 15 months of L-thyroxine treatment, there was a significant increase in free T4 and a significant decrease in TSH and antithyroglobulin antibody and antithyroid peroxidase antibody levels. CD8+ cell counts increased in both groups, CD4/CD8 levels decreased significantly because of the increase in CD8+ cell count levels. Though there was no change in cytological findings, ultrasonography showed a decrease in thyroid volume in L-thyroxine–receiving patients, whereas an increase was detected in patients who were followed without treatment." These results suggest that thyroid hormones may have some immune-modulating effects when used in a prophylactic manner before the individual becomes hypothyroid.[42]

Research has found that hypothyroidism does not lead to shifts of TH-1 or TH-2 cytokine dysregulation as a consequence of low thyroid hormone status. To summarize, it appears that thyroid hormones are used by immune cells to modulate active and functional expression of lymphocytes, but low thyroid hormone status is not directly related to abnormal cytokine surges or TH-1/TH-2 dominances.[43]

Thyroid Hormones

↓

Modulate CD4 and CD8 Activity

↓

Modulate TH-1 and TH-2 Expression

↓

Influence the Potential to Regulate Abnormal Autoimmune Expression

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

Hashimoto's Autoimmune Thyroid Attack

↓

Destroys Thyroid Gland

↓

Decreases Thyroid Hormone Production

↓

Inability to Modulate TH-1/TH-2 Activity

↓

Promotion and Flare-up of Hashimoto's Response

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

Immune Cell Apoptosis and Thyroid Hormones

"Apoptosis plays a critical role in the development of homeostasis of tissues, especially those with high cell turnover, such as the lymphoid system." Research has demonstrated that "T-lymphocytes treated with thyroid hormones accompany reduction of Bcl-2 protein expression, production of reactive oxygen species, and reduction of mitochondrial delta psi, resulting in apoptotic lymphocyte death."[44]

Summary of Thyroid Influence on Immunology

- Thyroid hormones produce a protective influence on the intestinal lining.
- Thyroid hormones increase the proliferation of intestinal IELs to improve the intestinal barrier system.
- Thyroid hormones modulate the expression of intestinal inflammation.
- Thyroid hormones increase intestinal catecholamines and enhance intestinal blood flow.
- Thyroid hormones modulate T- and B-cell mitogen-induced proliferation.
- Thyroid hormones enhance T-helper stimulation of nitric oxide synthase.
- Thyroid hormones enhance dendritic antigen-presenting-cell (APC) responses.
- Thyroid hormones modulate immune cell cytokine releases.
- Thyroid hormones modulate lymphocyte subpopulation in mediating inflammation.
- Thyroid hormones modulate dampening of excess inflammatory cytokine responses in an inflammatory state.
- Thyroid hormones modulate immune cell homeostasis apoptosis.

Part III: Immunology-Thyroid-Immunology Crosstalk

It is very clear that thyroid hormones are important for T-cell, B-cell, NK cell, and dendritic cell activity, proliferation, and modulation. Thyroid hormones exert numerous influences not only on proliferation of immune cells, but also on immune cell messenger release and immune cell homeostasis apoptosis. Lack of thyroid hormone exposure promotes intestinal permeability, exaggerated inflammatory responses, poor antigen recognition, and poor cytokine release modulation. It is important for the clinician to consider thyroid activity in immune-dysregulated or challenged patients.

On the other hand, increased immune activity promotes surges of cytokines that have been shown to dysregulate thyroid metabolism. Cytokines and adjuvants have the ability to dampen TRH, TSH, T4, and T3 release. They have been shown to promote abnormal coordination of the HPT axis, dysregulate T4-to-T3 conversion, and promote thyroid receptor-site transcription dysfunction. Thyroid disorders promoted by immunological activity termed NTI are very common in those suffering from chronic illnesses, chronic infections, and autoimmune responses. It is important for the clinician to understand these mechanisms to most effectively support physiology that can optimize thyroid function. In NTI cases, management of the underlying immune response may be more efficient in promoting thyroid responses than directly supporting the thyroid.

In chronic patients, it is not uncommon to find an overlap between the effects of the immune system and thyroid metabolism, and the effects of thyroid metabolism and immune function. Respect and consideration of these two systems is crucial in unwinding the abnormal crosstalk between them. Using combination support strategies to support thyroid function and immune modulation may be necessary in these who are chronically challenged.

Where a condition is identified or a lab result is outside the normal range: Patient should seek appropriate conventional medical management and drug administration by a physician.

Case Examples

Case 2

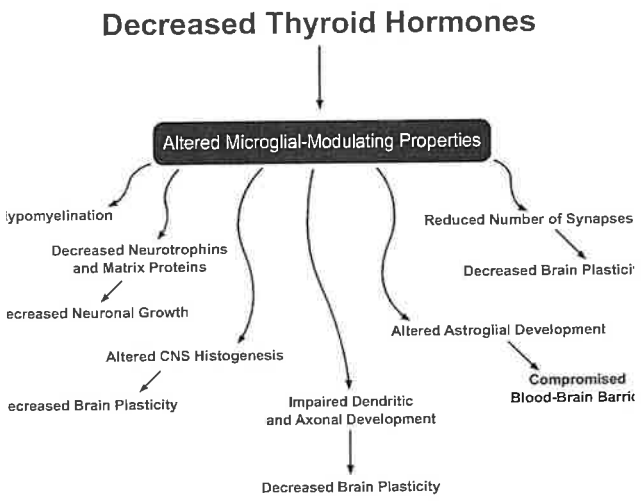
A patient has chronic hepatitis C and has started to notice recent thyroid symptoms. What mechanisms are possible?

Case 3

A recent blood test of a hypothyroid patient taking levothyroxine demonstrated elevated eosinophils. The patient also recently noticed that several of her thyroid symptoms returned. What mechanisms are possible?

Thyroid-Brain Crosstalk

The evidence clearly demonstrates that there is functional crosstalk between the brain and the thyroid. Thyroid hormones have very powerful influences on microglial cells and neuronal neurotransmitter pathways. This is no surprise, since individuals with thyroid disorders develop a vast list of neurological, mood, and cognitive symptoms. There is also clear evidence that neurotransmitters are important centrally for the modulation of the hypothalamus-pituitary-thyroid axis. In this section of the notes, a review of these interactions will be divided by interrelationships that include microglia, serotonin, acetylcholine, GABA, and dopamine. A summary of all these interrelationships will conclude this section on the thyroid-brain-thyroid connection.



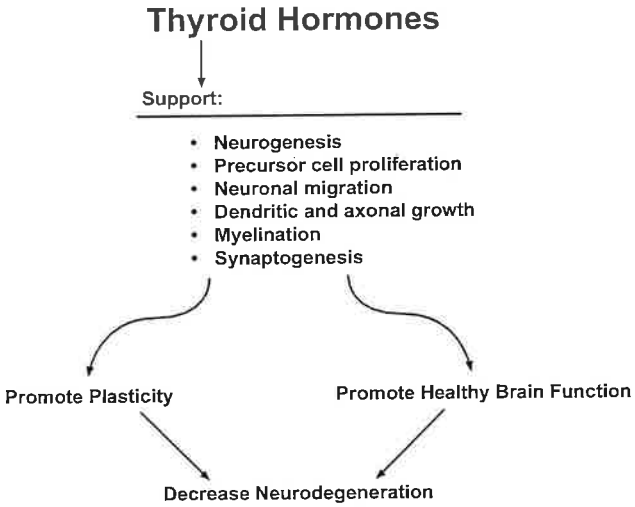
Microglia, Neurons, and Thyroid Interactions

Thyroid hormones have powerful microglia-modulating properties that are very important for brain development as well as healthy adult brain aging and neuroimmune modulation. Congenital hypothyroidism leads to developmental disorders characterized by cognitive, sensory, and motor disorders that are correlated with the magnitude of the thyroid hormone deficiency. This reflects a variety of alterations in CNS histogenesis, including impaired dendritic and axonal development, reduced number of synapses, hypomyelination, and altered astroglial development.[45-46]

Microglial cells in the developing brain produce neurotrophic factors such as neurotrophins and extracellular matrix proteins known to modulate neuronal growth. It is proposed that insufficient thyroid hormones leads to loss of microglial production of neurotrophic factors necessary for the establishment of quality neuronal networks.[47-49] Research has demonstrated that thyroid hormones are an important promoter of microglial growth and morphological differentiation.[50]

It has been well established that neuronal, microglial, and astroglial cells are recognized targets of thyroid hormones.[51-53] It is therefore possible that thyroid hormone deprivation might affect microglial growth indirectly by reducing the capacity of developing neurons or astrocytes to produce growth factors that promote survival, proliferation and/or differentiation of microglial cells.[53-56]

T3 and T4 thyroid hormones have been demonstrated to favor basic processes of neurogenesis, including precursor cell proliferation, neuronal migration, dendritic and axonal growth, myelination, and synaptogenesis.[55] In the adult brain, thyroid hormones have shown the ability to promote plasticity and promote healthy brain aging.[56] It is suggested that thyroid hormones might influence microglial reactions that accompany lesions or neurodegenerative diseases in the CNS and could regulate microglial functions.[50]



Summary of Thyroid Microglial Connections

- Thyroid hormones modulate microglial and neuronal function related to:
  - Production of neurotrophic factors

- Promotion of CNS histogenesis
- Promotion of microglial growth and morphological differences
- Production of matrix proteins to modulate neuronal growth
- Promotion of establishment of quality neuronal network
- Modulation of dendritic and axonal development
- Promotion of myelination
- Modulation of astroglial and blood-brain barrier development
- Modulation of synaptic activity
- Promotion of neuronal plasticity
- Modulation of microglia reactions to lesions and neurodegeneration

Serotonin and Thyroid Interactions

A study was conducted to evaluate the relationship between the hypothalamus-pituitary-thyroid (HPT) system and systemic serotonin by measuring platelet serotonin biochemical biomarkers (5-HT concentration and monoamine oxidase B). It was found that hypothyroid patients not treated with thyroid hormones had reduced systemic serotonin platelet biomarkers when compared with hypothyroid subjects who were on thyroid replacement therapy. This study suggests that thyroid hormones may have an influence on systemic serotonin activity.[57]

In a 2008 study, researches pointed out that "low cerebral bioenergetic metabolism has been reported in subjects with major depressive disorder (MDD). Thyroid hormones have been shown to increase brain bioenergetic metabolism." The researchers assess "whether changes in brain bioenergetics measured with phosphorus magnetic resonance spectroscopy (^{31}P MRS) correlate with treatment outcome during augmentation treatment with triiodothyronine (T3) in MDD." The study found that T3 augmentation produced significant changes in brain bioenergetic metabolism in treatment of MDD responders to thyroid hormone support.[58]

Thyroxine therapy has been shown to activate the 5-HT2A serotonin receptors in the animal frontal cortex.[59] A study was conducted to evaluate whether the addition of T3 helps relieve depressive symptoms in non-hypothyroid major depressive disorder patients who failed to respond to an adequate course of standard SSRI antidepressants. They found that 42% of the subjects who did not respond to SSRI therapy improved with T3 augmentation.[60]

A UCLA study was conducted "to examine the relation between baseline measurements of thyroid function and response to selective serotonin reuptake inhibitors (SSRI) and to consider the effect of these antidepressants on thyroid hormone levels." Subjects with major depression had measurements taken of TSH, T3, and T4 before and after treatment with a SSRI. The results found that "baseline TSH correlated strongly with response to treatment as measured by Hamilton Depression Rating Scale scores." Low TSH values correlated with greater improvement in depressive symptoms. They concluded by stating that, "baseline thyroid function, as measured by serum TSH, may predict a patient's response to antidepressant treatment with SSRIs. Optimal thyroid function, beyond simply being within the normal laboratory values, may be necessary for an optimal response to antidepressants."[61]

Serotonin and thyroid hormones are part of a complex system modulating energy expenditure. It has been found that serotonin is important in both central and peripheral thermoregulation, due to its site of action in modulating heat production in relation to controlling intracellular T3 production via the type II 5'-deiodinase enzyme.[62]

A review of the literature on the relationships between thyroid hormones and the brain serotonin (5-HT) system, limited to studies in adult humans and adult animals, has found intimate relationships between these two messengers. "In humans, neuroendocrine challenge studies in hypothyroid patients have shown a reduced 5-HT responsiveness that is reversible with thyroid replacement therapy. In adult animals with experimentally induced hypothyroid states, increased 5-HT turnover in the brainstem is consistently reported while decreased cortical 5-HT concentrations and 5-HT2A receptor density are less frequently observed. In the majority of studies, the effects of thyroid hormone administration in animals with experimentally induced hypothyroid states include an increase in cortical 5-HT concentrations and a desensitization of autoinhibitory 5-HT1A receptors in the raphe area, resulting in disinhibition of cortical and hippocampal 5-HT release. Furthermore, there is some indication that thyroid hormones may increase cortical 5-HT2 receptor sensitivity. In conclusion, there is robust evidence, particularly from animal studies, that the thyroid economy has a modulating impact on the

brain serotonin system. Thus, it is postulated that one mechanism, among others, through which exogenous thyroid hormones may exert their modulatory effects in affective illness is via an increase in serotonergic neurotransmission, specifically by reducing the sensitivity of 5-HT1A autoreceptors in the raphe area, and by increasing 5-HT2 receptor sensitivity."[63]

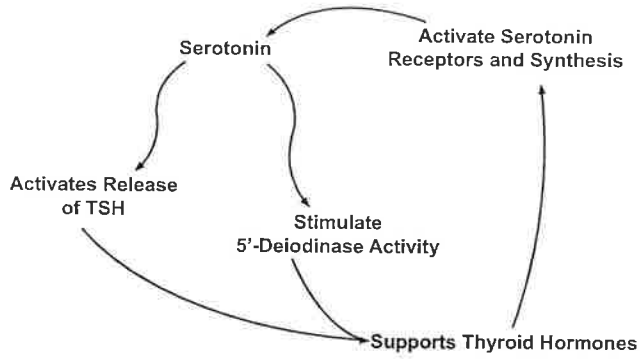
"The effects of hypothyroidism on 5-HTIA and 5-HT2A receptors and the serotonin transporter protein were studied in thyroidectomized male Wistar rats in two experimental groups: 1) animals kept on an iodine-free diet hypothyroid rats) and 2) animals kept on thyroxine (15 microg/kg) for 21 days (giving normal thyroid hormone levels. euthyroid animals) [sic]. Sham-operated rats served as controls. Binding of [3H]8-OH-DPAT with 5-HTIA receptors and [3H]citalopram with the transporter protein in the hippocampus and midbrain showed no changes in hypothyroid rats as compared with controls. Conversely, there were significant decreases in [3H]ketanserin binding to 5-HT2A receptors in the frontal cortex in hypothyroid rats as compared with controls; this decrease was reversed by thyroxine treatment. Thus, losses of cortical 5-HT2A receptors appear to be the main consequence of hypothyroidism at the level of the serotonin system of the brain."[64]

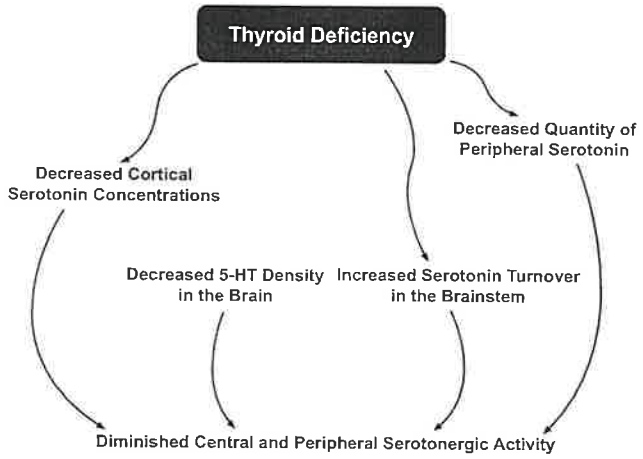
Effects of thyroid hormone deficiency on 5-HT1A receptors, 5-HT2A receptors, and serotonin transporter in the brain were studied in thyroidectomized Wistar rats receiving an iodine-free diet and receiving 15 micrograms/kg of thyroxine for 21 days. Binding of 3H-8-OH-DPAT to 5-HT1A receptors and 3H-cytalopram to serotonin transporter were unchanged in hypothyroid rats as compared to the control. 3H-ketanserin binding to 5-HT2A receptors was significantly decreased in the frontal cortex in hypothyroid rats. The cortical 3H-ketanserin binding in thyroidectomized rats was normalized after thyroxine replacement. The "data suggest that the decrease in the cortical 5-HT2A receptors is the main consequence of impairing effect of hypothyroidism on serotonin neurotransmission."[65]

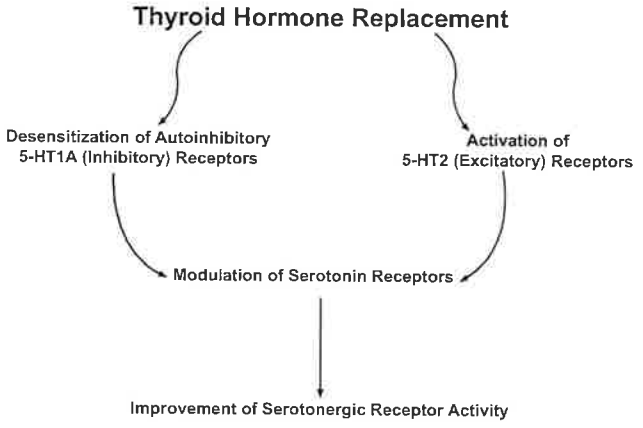
"Although a link between the hypothalamic-pituitary-thyroid (HPT) axis and affective disorder has been established the mechanism underlying this relationship remains unclear. Since the serotonin (5-HT) system appears to be involved in the pathophysiology of mood disorders, the time course of the effects of thyroidectomy (TXT) with or without thyroxine (T4) replacement on 5-HT1A receptors and 5-HT uptake sites was examined. TXT caused a significant increase in 3H-8-hydroxy-2-(di-n-propylamino)-tetralin (3H-DPAT) binding to 5-HT1A receptors in the cortex and hippocampus at 7 days, and this increase was also evident at 35 days following TXT. Administration of high-dose T4 for 28 days caused the binding of 3H-DPAT to recover to sham levels in the cortex, to increase in the hippocampus and hypothalamus, and had no effect in the dorsal raphe nucleus. Replacement with high-dose T4 had no effect on 3H-CN-IMI binding to 5-HT uptake sites when compared to sham-operated animals at all time points examined. These results suggest that a neuromodulatory link may exist between the HPT axis and 5-HT1A receptors in the limbic regions of the rat brain. Depending on the brain region examined, a differential response to circulating levels of thyroid hormone was observed."[66]

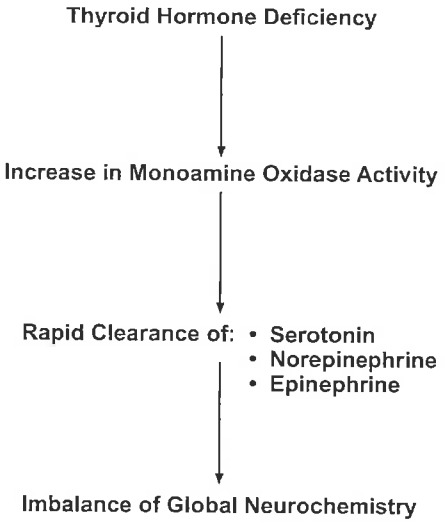
Summary of Serotonin and Thyroid Interactions

- Thyroid hormones have been shown to have a direct influence on the quantity of system peripheral serotonin measurements.
- Thyroid hormones have been shown to increase bioenergetic metabolism in patients with MDD who were responders to thyroid hormones.
- Thyroid hormone therapy has been shown to activate 5-HT2A serotonin receptors in the brain.
- Thyroid deficiency leads to increased serotonin turnover in the brainstem.
- Thyroid deficiency leads to decreased cortical serotonin concentrations.
- Thyroid deficiency has been shown to decrease 5-HTA density in the brain.
- Serotonin is used as a messenger to release TSH from the pituitary.
- Serotonin is involved with modulating intracellular T3 production via type II 5'-deiodinase
- Administration of thyroid hormones in hypothyroid subjects has been shown to increase cortical serotonin concentrations and cause a desensitization of autoinhibitory of 5-HT1A (inhibitory) receptors an increase in 5-HT2 (excitatory) receptor sensitivity





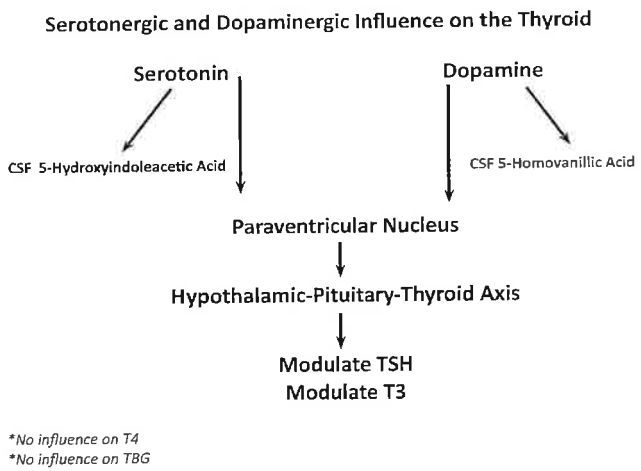




Serotonin Influences on TSH Release

5-HT has modulating influences on the HPT axis and the regulation of TSH.67-68 However, consistent exposure to 5-HT appears to promote cellular insensitivity of thyroid cells.69 Dopamine has modulating influences on the HPT axis and the coordination of TSH release due to its inhibitory influences.70 There is an intermingled relationship between TSH and dopamine, and dopamine and TSH, which coordinate appropriate TSH output.[71-73]

The neurotransmitters serotonin and dopamine exert some influence on the hypothalamus-pituitary-thyroid axis. A study examined simultaneous samples of serum thyroid hormones (TSH, total T4, free T4, total T3, free T3) and cerebrospinal measurements of homovanillic acid and 5-hydroxyindoleacetic acid, the major metabolites of dopamine and serotonin. "CSF homovanillic acid concentrations were significantly and negatively correlated with plasma TSH and T3 (free and total), but not with T4 (free or total). CSF 5-hydroxyindolacetic acid levels were significantly and negatively correlated with plasma TSH and total T3, but not with free T3 or T4 (free or total)."[74] Therefore, the findings suggest that these monoamine neurotransmitters found in the central nervous system exert influence on HPT axis activity, but not on thyroid-binding-globulin states. Their influence was limited to TSH and T3, but not T4. Therefore, mechanisms distinctive in the thyroid gland may contribute to T4 regulation unrelated to central influence of serotonin and dopamine.



Acetylcholine and Thyroid Interactions

A study was conducted "to investigate how changes in metabolism induced by thyroid hormones could affect the activities of acetylcholinesterase (AChE), (Na^+,K^+)-, and Mg^{2+}- adenosinetriphosphatase (ATPase) in the frontal cortex and the hippocampus of adult rats." "In hypothyroidism, AChE activity was significantly decreased in the frontal cortex (-23%) and increased in the hippocampus (+21%). Na^+,K^+-ATPase activity was significantly decreased in both the frontal cortex (-35%) and the hippocampus (-43%) of hypothyroid rats." In conclusion, thyroid hormones are used to modulate acetylcholine activity. Lack of thyroid hormones leads to diminished AChE activity and increased acetylcholine release in order to enhance cholinergic activity.[75] Thyroid hormone deficiency may lead to acetylcholine depletion and inefficient cholinergic synapses.

Another study was conducted "to investigate how changes in the metabolism induced by thyroid hormones could affect the activities of AChE, (Na^+, K^-)- and Mg^{2+}-ATPase in the hypothalamus and the cerebellum of adult rats." They found that decreased thyroid hormone levels lead to decreased Na^+,K^+-ATPase activity and increased synaptic acetylcholine release, and resulted in a decrease in the AChE activity of the cerebellum.[76]

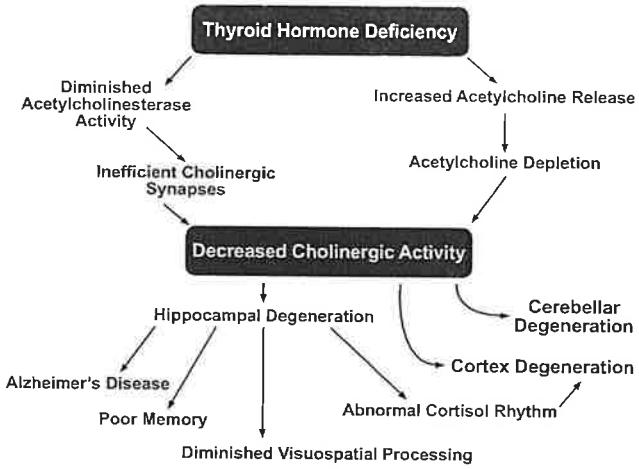
"In addition to their role in cellular metabolic activity, thyroid hormones also regulate neural development; the central nervous system is particularly dependent on thyroid hormones for normal maturation and function. Specifically, there appears to be extensive inter-reliance between thyroid hormones and acetylcholine, nerve growth factor, and hippocampal function." Research has found that lack of thyroid hormones leads to diminished activities in these systems and impacts performance of spatial learning tasks.[77]

Triiodothyronine has been found to have direct influence on acetylcholine metabolism. One study evaluated the influence of thyroid hormones in cholinergic systems in the cerebral cortex. They found that administration of L-triiodothyronine in single doses increased the synaptosomal acetylcholinesterase and Mg^{2+}-ATPase activity maximally at 24 hours in a dose-dependent way. The results demonstrate that T3 stimulates acetylcholine and there is a direct influence of thyroid hormones on cholinergic pathways.[78]

A study was conducted to evaluate the influence of hypothyroidism in the adult rat on brain biochemistry. "Hypothyroidism was induced in 6-month-old male rats by partial thyroidectomy coupled with the administration of 6-n-propyl-thiouracil in the drinking water." The results found "acetylcholine esterase activity was reduced in the cerebellum (by 45%), the medulla (by 34%), and the subcortex (by 45%), whereas monoamine oxidase activity was affected in only one region, the cerebellum, where it was increased (by 61%)." The results of hypothyroidism lead to global loss of cholinergic activity, but with greatest potential to develop cerebellum degeneration.[79] Research on animal studies evaluating the effects of thyroid hormones on cholinergic transmission found that T4 increases cholinergic transmission.[80]

Summary of Acetylcholine Thyroid Interactions

- Thyroid hormone deficiency leads to diminished acetylcholinesterase activity and increased acetylcholine release in an attempt to increase cholinergic activity.
- Thyroid hormone deficiency may lead to acetylcholine depletion and inefficient cholinergic synapse.
- Thyroid hormones increase synaptosomal acetylcholinesterase activity.
- Thyroid hormone deficiency leads to increased monamine oxidase activity with global loss of cholinergic activity with greatest potential to develop cerebellar degenerations.
- Thyroid hormones increase cholinergic transmission.



For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

GABA and Thyroid Hormone Interactions

"Thyroid hormones have critical roles in brain development and normal brain function in vertebrates. Clinical evidence suggests that some human nervous disorders involving GABA (gamma-aminobutyric acid)-ergic systems are related to thyroid dysfunction (ie, hyperthyroidism or hypothyroidism). There is experimental evidence from in vivo and in vitro studies on rats and mice indicating that thyroid hormones have effects on multiple components of the GABA system. These include effects on enzyme activities responsible for synthesis and degradation of GABA, levels of glutamate and GABA, GABA release and reuptake, and GABA(A) receptor expression and function. In [the] developing brain, hypothyroidism generally decreases enzyme activities and GABA levels, whereas in [the] adult brain, hypothyroidism generally increases enzyme activities and GABA levels. Hyperthyroidism does not always have the opposite effect. In vitro studies on [the] adult brain have shown that thyroid hormones enhance GABA release and inhibit GABA-reuptake by rapid, extranuclear actions, suggesting that the presence of thyroid hormones in the synapse could prolong the action of GABA after release. There are conflicting results on effects of long-term changes in thyroid hormone levels on GABA reuptake. Increasing and decreasing circulating thyroid hormone levels experimentally in vivo alter density of GABA(A) receptor-binding sites for GABA and benzodiazepines in [the] brain, but results vary from study to study, which may reflect important regional differences in the brain. There is substantial evidence that thyroid hormones also have an extranuclear effect to inhibit GABA-stimulated Cl(-) currents by a non-competitive mechanism in vitro. The thyroid gland exhibits GABA transport mechanisms as well as enzyme activities for GABA synthesis and degradation, all of which are sensitive to thyroidal states. In rats and humans, GABA inhibits thyroid stimulating hormone (TSH) release from the pituitary, possibly by action directly on the pituitary or on hypothalamic thyrotropin-releasing hormone neurons. In mice, GABA inhibits TSH-stimulated thyroid hormone release from the thyroid gland. Taken together, these studies provide strong support for the hypothesis that there is reciprocal regulation of the thyroid and GABA systems in vertebrates."[81]

"The effects of a mild hypothyroidism condition on benzodiazepine (BDZ) and mu-opioid receptor levels were investigated. Female Wistar rats were randomly divided into two groups: 1) hypothyroid rats, which received methimazole in drinking water for four weeks, and 2) euthyroid rats, which drank only tap water. Animals were sacrificed and their brains were used for autoradiography experiments. When compared to the euthyroid group, the hypothyroid group presented reduced benzodiazepine receptor binding in medial amygdala (24%) and high mu-receptor levels in frontal (25%), sensorimotor (65%) and temporal (29%) cortices, basolateral amygdala (50%) and ventroposterior thalamic nucleus (49%). The present data suggest that alterations in BDZ and mu-receptor binding could be associated with the higher excitability observed in animals with triiodothyronine (T(3)) deficiency."[82]

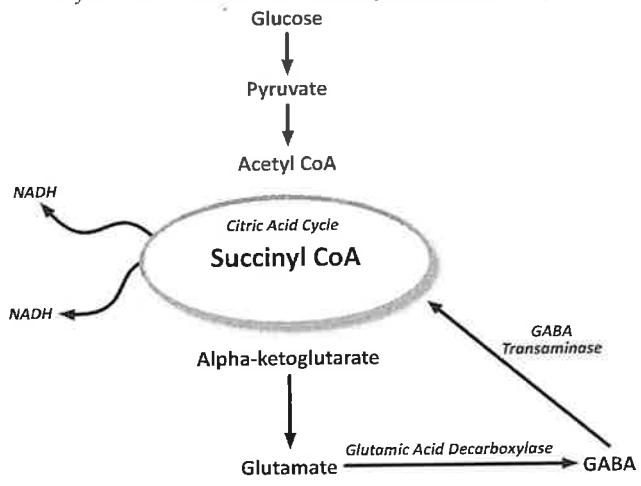
"The effect of administration of thyroid hormones on central benzodiazepine receptors was investigated using neuron-enriched primary cultures obtained from the neopallium of 16-day-old embryonic rats. Addition of L-triiodothyronine for 3 days decreased the maximal number of benzodiazepine receptor binding sites without any change in affinity at 10(-5) and 10(-6) M. L-Thyroxine administered for 3 days had the same effect at 10(-5) M." "The decrease in number of cerebral cortical neuronal benzodiazepine receptors due to L-isomers of thyroid hormones may be related to the convulsions and anxiety observed in thyrotoxicosis in humans."[83]

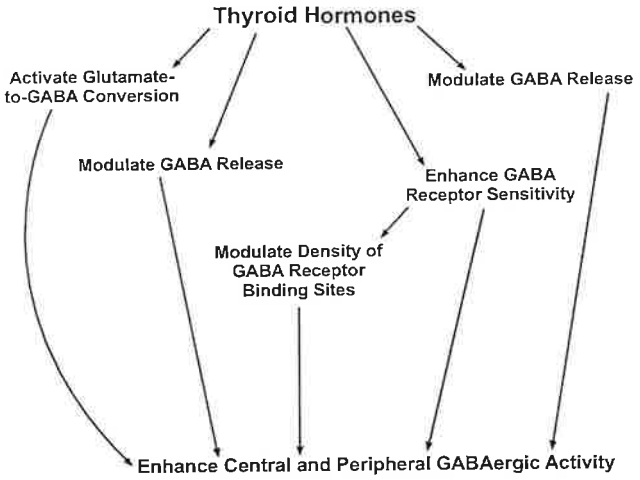
"Studies of the effects of thyroid hormones on the uptake of neurotransmitters by homogenates of rat

cerebral cortex have revealed a significant competitive inhibition of neuronal uptake of [3H]GABA by thyroid hormones (T3 greater than T4 greater than reverse T3)." "GABA uptake in homogenates of cerebral cortex from hypothyroid rats was significantly enhanced over that of controls; however, uptake in tissues from hyperthyroid rats was not significantly diminished."[84]

Summary of GABA and Thyroid Interactions

- Thyroid hormones have effects on enzyme activities responsible for synthesis and degradation of GABA.
- Thyroid hormones have effects on the levels of glutamate to GABA conversion.
- Thyroid hormones have effects on GABA release and reuptake.
- Thyroid hormones have influences on GABA(A) receptor expression.
- In developing brains, hypothyroidism generally decreases enzyme activities and GABA levels.
- In adult brains, hypothyroidism generally increases enzyme activities and GABA levels.
- Hyperthyroidism does not always have opposite effects on the GABA system.
- Thyroid hormones enhance GABA release and inhibit GABA reuptake by rapid extranuclear actions.
- Thyroid hormones prolong the action of GABA in the synapse after it is released.
- Increasing and decreasing circulating thyroid hormone levels experimentally in vivo alter density of GABA(A) receptor-binding sites for GABA and benzodiazepines in the brain.
- GABA is used to modulate inhibition of TSH release in the pituitary.
- GABA modulates inhibition of TSH-stimulated thyroid hormones from the thyroid gland.





Dopamine and Thyroid Metabolism

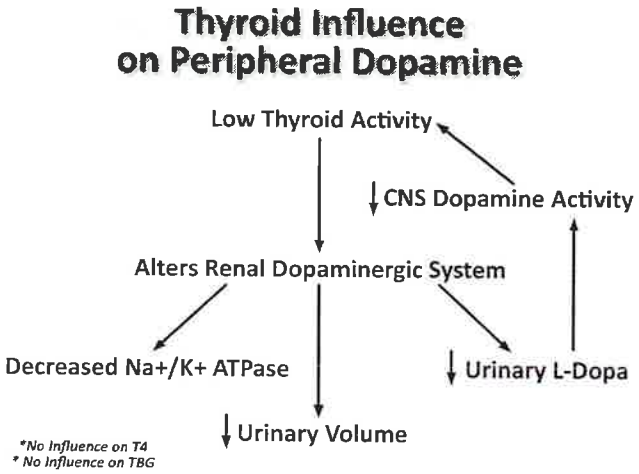
"Adult-onset hypothyroidism is associated with neurological changes such as cognitive dysfunction and impaired learning, which may be related to alterations of synaptic plasticity." Researchers investigated "the consequence of adult-onset hypothyroidism on thyroid-mediated transcription events in striatal synaptic plasticity, and the effect of triiodothyronine (T3) replacement." They "used hypothyroid mice treated with propylthiouracil (PTU) and methimazole (MMI), with or without subsequent administration of T3." They "evaluated the amount of T3 nuclear receptors (TR-alpha-1, TR-beta) and striatal plasticity indicators: neurogranin (RC3), Ras homolog enriched in striatum (Rhes), Ca2+/calmodulin-dependent protein kinase (CaMKII), and dopamine- and cAMP-regulated phosphoprotein (DARPP-32)." They also "assessed hypothyroid mice motor behavior as related to striatum synaptic functions. Hypothyroid mice exhibited significantly reduced TRbeta, RC3, and Rhes expression. T3 administration reversed the expression of TRbeta, RC3, and up-regulated CaMKII levels as well as motor behavior, and decreased DARPP-32 protein phosphorylation." They "suggested that thyroid hormone modulation had a major impact on striatal synaptic plasticity of adult mice, which produced in turn motor behavior modifications."[85]

"Microdialysis cannulas were surgically implanted into the striatum of thyroparathyroidectomized (TX) and sham rats in order to determine differences in dopamine release and core body temperature following MDMA administration. Rats were subsequently treated with

MDMA (10 mg/kg, s.c.), and striatal DA levels were monitored at 20-minute intervals, as well as core temperature every 30 min. Sham rats responded to MDMA with a characteristic hyperthermic response and significant increases in extracellular dopamine. Conversely, TX rats responded to MDMA with a hypothermic response and failed to demonstrate a similar increase from basal dopamine levels. On the basis of these data, thyroid hormones are not only important in the thermogenic effects of MDMA but also appear to have an auxiliary role in MDMA-induced striatal dopamine release."[86]

Thyroid Influences on Peripheral Dopamine

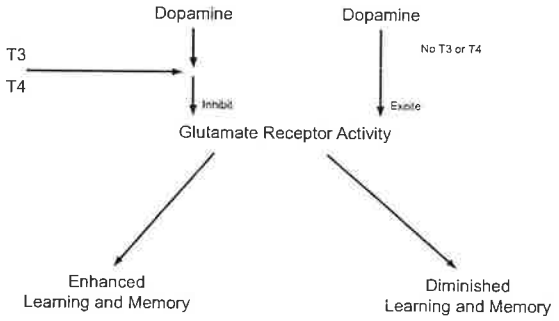
Thyroid hormones have strong influences on the renal dopaminergic system, and the response of dopamine receptors are altered by thyroid hormone deficiencies. A recent study incorporating surgical thyroidectomy in animals and treatment with the thyroid-inhibiting drug "2-thiouracil (Thio) decreased renal cortex Na+/K+ ATPase activity and urinary volume. Thyroidectomy also decreased urinary Na+ and urinary L-DOPA without changing urinary excretion of dopamine."[87] These findings suggest that thyroid hormones play a significant role in urinary volume, electrolyte balance, and urinary elimination of waste by its influences on the peripheral (renal) dopaminergic system. Therefore, failure in thyroid hormones, dopamine activity, or dopamine receptor responsiveness may hinder the body's urinary elimination process.



Thyroid Influences on Central Nervous System Dopamine

In the central nervous system, dopaminergic D(2) receptors inhibit the glutamatergic (excitatory) transmission in the hippocampus and therefore provide dampening influences on the hippocampus. Thyroid hormones appear to positively influence the inhibitory influence of dopamine on glutamatergic receptor activity. A recent animal study found that conditions of induced thyroid deficiency prevented the inhibitory influences of the dopaminergic D(2) receptor on glutamatergic activity.[88] These findings present a possible mechanism by which a lack of thyroid hormones may promote deficits in central nervous system activity such as memory and cognition.

Thyroid and CNS Dopamine Response



Hypothyroidism and Dopamine Release

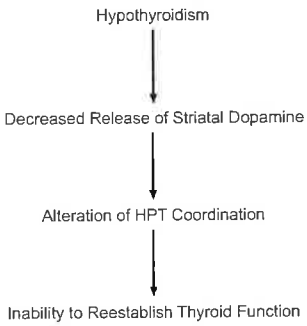
Hypothyroidism has been shown to promote a loss of striatal dopamine release. In a recent animal study, animals were either placed in a control group or underwent thyroidectomy. Both groups had surgically implanted into the striatum microdialysis cannulas in order to determine differences in dopamine release and core body temperature following 3,4-methylenedioxymethamphetamine (MDMA or Ecstasy) administration. MDMA was used due to its powerful stimulus on striatal dopamine release. The outcome demonstrated that subjects that had thyroidectomy did not produce dopamine levels comparable to the control subjects.[86] This data suggests that thyroid hormones play an important role in the central output of dopamine.

A study that examined the levels of monamine precursors, neurotransmitters, and their metabolites in the cerebrospinal fluid of newly diagnosed hypothyroid patients were evaluated before and after thyroid replacement therapy. Thyroid therapy appeared to significantly decrease the levels of dopamine and the noradrenaline metabolite, 4-hydroxy-3-

methoxyphenylglycol (HMPG). The researches concluded that "there is an interaction between thyroid function and CSF disposition of monoamine compounds."[89]

Associations between dopamine regulation of TSH and Parkinson's disease have been reported in the literature. Thyroid hormones appear to have some influence on Parkinsonian tremor, but its role in the management of Parkinson's disease is not clear or understood.[90]

Hypothyroidism and Dopamine Release

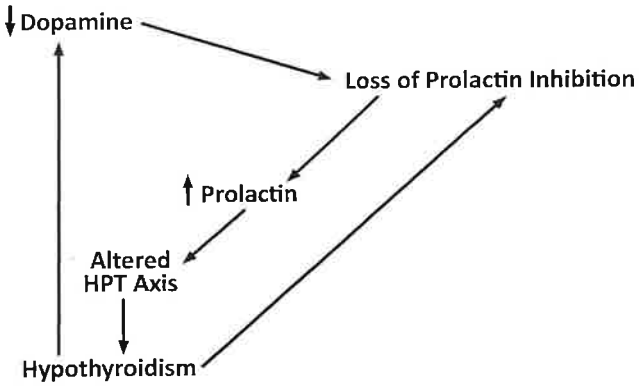


For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

Thyroid-Dopamine-Prolactin Loop

Dopamine, prolactin, and TRH/TSH coordination all appear to have a closed-loop physiological relationship with each other.[91] Decreased dopaminergic activity induced by the medication domperidone enhanced TRH-induced TSH release by 50%.[92] Therefore, reduced dopamine levels may promote exaggerated TSH output upon TRH stimulation and alter normal hypothalamus-pituitary coordination. Additionally, diminished dopaminergic activity may lead to a loss of prolactin inhibition that alters hypothalamic TRH regulation of the pituitary TSH release.[93] On the other hand, hyperprolactinemia may be secondary to hypothyroidism and thyroxine therapy may normalize the elevated prolactin.[94]

Thyroid-Dopamine-Prolactin Loop

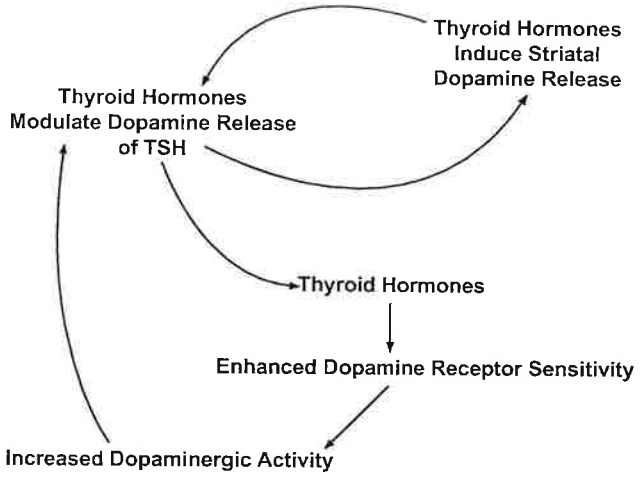


Hyperthyroidism and Dopamine

Hyperthyroid states decrease the concentration of dopamine in the anterior, middle hypothalamus, and the preoptic area of the central nervous system.[95]

Summary of Dopamine and Thyroid

- Dopamine modulates dopamine release of TSH
- Thyroid hormones induce striatal release
- Thyroid hormones modulate striatal synaptic plasticity
- Thyroid hormones modulate the renal dopaminergic system
- Thyroid hormones modulate excitation or inhibition of heterotrophic modulation of glutamate receptors in the hippocampus for learning
- Deficiency of thyroid hormones leads to decreased brain dopamine synthesis
- Thyroid hormones lead to decreased adrenal and peripheral dopamine synthesis
- Thyroid hormones help modulate Parkinsonian tremor [99]



Summary of Thyroid-Brain Crosstalk

- Thyroid hormones modulate microglial and neuronal function related to:
  - Production of neurotrophic factors
  - Promotion of CNS histogenesis
  - Promotion of microglial growth and morphological differences
  - Production of matrix proteins to modulate neuronal growth
  - Promotion of establishment of quality neuronal network
  - Modulating dendritic and axonal development
  - Promotion of myelination
  - Modulating astroglial and blood-brain barrier development
  - Modulating synaptic activity
  - Promotion of neuronal plasticity
- Thyroid hormones modulate microglial reactions to lesions and neurodegeneration.
- Thyroid hormone deficiency leads to diminished acetylcholinesterase activity and increased acetylcholine release in an attempt to increase cholinergic activity.
- Thyroid hormone deficiency may lead to acetylcholine depletion and inefficient cholinergic synapse.
- Thyroid hormones increase synaptosomal acetylcholinesterase activity.
- Thyroid hormone deficiency leads to increased monoamine oxidase activity with global loss of cholinergic activity with greatest potential to develop cerebellum degenerations.
- Thyroid hormones increase cholinergic transmission.
- Thyroid hormones have been shown to have a direct influence on the quantity of system peripheral serotonin measurements.
- Thyroid hormones have been shown to increase bioenergetic metabolism in patients with major depressive disorder who were responders to thyroid hormones.
- Thyroid hormone therapy has been shown to activate 5-HT2A serotonin receptors in the brain.
- Thyroid deficiency leads to increased serotonin turnover in the brainstem.
- Thyroid deficiency leads to decreased cortical serotonin concentrations.
- Thyroid deficiency has been shown to decrease 5-HTA density in the brain.
- Serotonin is used as a messenger to release TSH from the pituitary.
- Serotonin is involved with modulating intracellular T3 production via type II 5'-deiodinase.
- Administration of thyroid hormones in hypothyroid subjects has been shown to increase cortical serotonin concentrations, a desensitization of autoinhibitory of 5-HT1A (inhibitory) receptors, and an increase in 5-HT2 (excitatory) receptor sensitivity.
- Thyroid hormones have effects on enzyme activities responsible for synthesis and degradation of GABA.
- Thyroid hormones have effects on the levels of glutamate to GABA conversion.
- Thyroid hormones have effects on GABA release and reuptake.
- Thyroid hormones have influences on GABA(A) receptor expression.
- In developing brains, hypothyroidism generally decreases enzyme activities and GABA levels.
- In adult brains, hypothyroidism generally increases enzyme activities and GABA levels.
- Hyperthyroidism does not always have opposite effects on the GABA system.
- Thyroid hormones enhance GABA release and inhibit GABA reuptake by rapid extranuclear actions.
- Thyroid hormones prolong the action of GABA in the synapse after it is released.
- Increasing and decreasing circulating thyroid hormone levels experimentally in vivo alter density of GABA(A) receptor-binding sites for GABA and benzodiazepines in the brain.
- GABA is used to modulate inhibition of TSH release in the pituitary gland.
- GABA modulates the inhibition of TSH stimulated thyroid hormones from the thyroid gland.
- Dopamine modulates dopamine release of TSH.

- Thyroid hormones induce striatal dopamine release.
- Thyroid hormones modulate striatal synaptic plasticity.
- Thyroid hormones modulate renal dopaminergic system.
- Thyroid hormones modulate excitation or inhibition of heterotrophic modulation of glutamate receptors in the hippocampus for learning.
- Deficiency of thyroid hormones leads to decreased brain dopamine synthesis.
- Thyroid hormones lead to decreased adrenal and peripheral dopamine synthesis.
- Thyroid hormones help modulate Parkinsonian tremor.[99]

Where a condition is identified or a lab result is outside the normal range: Patient should seek appropriate conventional medical management and drug administration by a physician.

Case Examples

Case 4

A new patient who has hypothyroidism tells you that she has felt better since taking thyroid hormones the past couple of years, but her memory is failing and she has difficulty thinking clearly at times. What mechanisms are suggested?

Case 5

A patient comes in with a history of taking thyroid hormones for several years. She tells you that she has been struggling with depression for as long. When she first tried SSRI antidepressants, they helped a little bit for the first few weeks but stopped being beneficial after a couple of months. She has tried several different types of thyroid hormones, but none has made a significant difference. What mechanisms must be considered?

Case 6

A patient has had hypothyroidism for several years and has struggled with depression. The only antidepressant that has worked is Abilify®. The patient noticed immediate improvements with the medication, but after the first year, the symptoms of depression returned. Her TSH has not changed in the past year. The patient wants your advice about increasing or changing her thyroid hormones. What do you think?

Case 7

A patient suffering with rheumatoid arthritis and hypothyroidism has been taking Synthroid® and hydrocortisone for the past seven years. The past few months, she has gotten lost driving home. She also cannot remember names of people when she sees them, and sometimes she even forgets who they are. What mechanisms must be considered?

Case 8

A new patient comes into your office currently taking Xanax® and Synthroid®. She tells you that her anxiety attacks have gotten much worse in the past six months. She also tells you that her anxiety attacks became more frequent when she stopped her carbohydrate-restricted diet. She recently had laboratory testing completed, and she had a fasting glucose of 180. She is confused, because she is not overweight and has been watching her diet and exercising regularly. What mechanisms must be considered?

Please see registered trademark ownership in Appendix 1.

Thyroid-Endocrine Crosstalk

There are clear endocrine interactions between the thyroid, pituitary-gonadal pathways, pituitary-ovarian pathways, reproductive hormone receptor expressions and sensitivities, reproductive gland enzyme activities, glycemic control, hormone binding ratios, hypothalamic modulation, etc. An understanding of these interrelationships is important for understanding the diverse set of symptoms and clinical presentations that accompany thyroid disorders. These interactions are very important to recognize because many times they are secondary to the thyroid, but the patient has silent thyroid presentations. For example, a patient may present with infertility as the chief complaint, with no thyroid symptoms, but it is her thyroid disorder that is creating the imbalances of the pituitary-ovarian feedback loop leading to infertility. Part I of this section will review the impacts of thyroid hormones on other endocrine metabolic pathways.

Part II of this section will review how other hormones influence thyroid metabolism. Reproductive hormones and cortisol have the potential to alter the pituitary-thyroid axis, alter the percentage of thyroid-binding proteins, impact thyroid hormone synthesis, down-regulate peripheral hormone conversion, and promote thyroid receptor-site resistance. A review of these patterns will be presented in this section and then clinically applied in the subsequent sections of this manual.

This section will end with Part III, which reviews the thyroid-endocrine-thyroid interplay. The clinician must be able to determine the ongoing influences of thyroid hormones on the endocrine system and how endocrine hormones influence thyroid metabolism. A clear understanding of these interrelationships leads to proper thyroid evaluation and management that is so necessary with the complex metabolic cases that we all face today.

Part I: How Thyroid Activity Influences the Endocrine System

Ovarian Cysts and the Thyroid

"Ovarian cysts have been reported in girls with long-standing uncompensated primary hypothyroidism. Restoration of a euthyroid state has been associated with the resolution of these cysts." A study evaluated the outcome in ten girls with ovarian cysts and hypothyroidism managed at [a] hospital with special emphasis on subsequent pubertal development and ovarian imaging." "Thyroxine replacement led to normalization of TSH levels with reversal of pubertal changes and regression of ovarian cysts in all patients 2.2 +/- 1.0 months after treatment. At last follow-up 3.5 +/- 2.6 years after initiation of treatment at the age of 12.0 +/- 2.3 years, all patients had normal ovarian size in ultrasound evaluation with six girls progressing to normal puberty." The study "emphasizes the need to exclude hypothyroidism in young girls with ovarian cysts."[96]

Another study was conducted to "assess the role of the functional states of pituitary-ovary, adrenal, and thyroid axes in the pathophysiology of polycystic ovarian syndrome (PCOS)." The researchers found, through "TRH testing, exaggerated thyroid-stimulating hormone and prolactin responses in PCOS groups and blunted free thyroxine response." "Enhanced adrenal activity may contribute to both hyperandrogenism and insulin resistance in this syndrome, and subclinical hypothyroidism may exist."[97]

Whenever ovarian cysts are identified, additional evaluation in regards to insulin surging disorders such as hypoglycemia and insulin resistance should be conducted. Insulin surges upregulate the activity of the enzyme 17,20 lyase and promote cyst development and androgen excess. This would especially be necessary if the patient presents with a history of dysglcyemia. However, if dysglycemia is not present and the patient is not overweight, careful attention to her thyroid needs to be considered. The conventional healthcare system will typically not evaluate the thyroid when ovarian cysts are identified and simply place the patient on oral contraceptives to shrink their cysts and falsely normalize their menstrual pattern. If the mechanisms of the cysts are thyroid-related, proper management of the thyroid will resolve the cysts.

Reproductive Hormones and the Thyroid

A study was "undertaken to examine the systemic role of thyroid hormones by analyzing changes in reproductive functions in hypothyroid female rats. Serum concentrations of triiodothyronine (T3) significantly decreased 1 week after the initiation of propyl-thiouracil treatment [thyroid-suppressing drug] or thyroidectomy. The estrous cycle became irregular 3 and 2 weeks

after the initiation of propyl-thiouracil treatment and thyroidectomy, respectively. Serum luteinizing hormone (LH) levels significantly reduced in both groups on the day of diestrus-1 about 1 month later. Hypothyroid rat shows the high progesterone and low testosterone levels. No significant changes in inhibin and estradiol levels were detected. The serum levels of FSH decreased in the thyroidectomy group. The irregular estrous cycle and the above changes in hormone levels were recovered by administration of T4. Compensatory secretions of FSH and LH induced by ovariectomy were enhanced by thyroidectomy and suppressed by T4 treatment. These results suggest that thyroid hormones play a role in the regulation of reproductive hormones secretion in the cyclic ovary."[98]

Whenever abnormal or dysregulated FSH and LH patterns are recognized on a female hormone profile, and especially when progesterone is elevated with depressed LH not related to exogenous progesterone, thyroid dysfunction needs to be evaluated. Additionally, many times when hypothyroidism is present, progesterone resistance will result, causing the progesterone receptor sites to lose their responsiveness to progesterone and progesterone levels will elevate. These individuals will present with a complex set of symptoms that may be specific to their menstrual disorders and not their thyroid, so it is necessary for the culprit, the thyroid, to be evaluated.

Male Testicular Cell Activity and the Thyroid

"Leydig cell steroidogenic activity under basal and stimulated conditions was studied in hypothyroid rats." In a study, "hypothyroidism was induced at a prepubertal age (30 days postpartum) by surgical thyroidectomy, and L-thyroxine (T4) supplementation (6 micrograms/100 g body weight/day for 30 days) to hypothyroid rats was begun after 30 days. Hypothyroidism for 60 days reduced serum LH and FSH without affecting prolactin. Serum and intratesticular testosterone and the specific activity of Leydig cell 3 beta- and 17 beta-hydroxysteroid dehydrogenases diminished in hypothyroid rats. The stimulatory effect of LH on Leydig cell steroidogenic activity and cAMP was also adversely affected in hypothyroid rats. All these changes were reversed by T4 supplementation. The present results of the study suggest that prepubertal hypothyroidism suppresses both basal and stimulated Leydig cell activity in adult rats."[99]

"Thyroid hormones are important for the growth and development of many tissues. Altered thyroid hormone status causes testicular abnormalities. For instance, juvenile hypothyroidism/neonatal transient hypothyroidism induces macroorchidism, increases testicular cell number (Sertoli, Leydig, and germ cells) and daily sperm production. Triiodothyronine (T3) receptors have been identified in sperm, developing germ cells, Sertoli, Leydig, and peritubular cells. T3 stimulates Sertoli cell lactate secretion as well as mRNA expression of inhibin-alpha, androgen receptor, IGF-I, and IGFBP-4. It also inhibits Sertoli cell mRNA expression of Müllerian inhibiting substance (MIS), aromatase, estradiol receptor, and androgen binding protein (ABP) and ABP secretion. T3 directly increases Leydig cell LH receptor numbers and mRNA levels of steroidogenic enzymes and steroidogenic acute regulatory protein. It stimulates basal and LH-induced secretion of progesterone, testosterone, and estradiol by Leydig cells. Steroidogenic factor-1 acts as a mediator for T3-induced Leydig cell steroidogenesis. "It is clear that T3 directly regulates Sertoli and Leydig cell functions."[100]

Although hypothyroidism is not common in men, it does occur and it typically leads to severe alterations in male hormone profiles. The impacts of thyroid hormones on male androgenic pathways are so diverse that potentially any abnormal pattern could be secondary to thyroid dysfunction. These patterns become difficult to evaluate as a thyroid problem since the male's dysfunctional androgen disorders are the presenting symptomatology. Thyroid hormones can directly cause abnormal male hormone imbalances. On the other hand, when Hashimoto's is identified in a male, it is common to see antibodies against the testicular cells, testosterone, and/or sperm. Therefore, the connection with abnormal male hormone patterns and thyroid can be directly related to abnormal thyroid hormone influences, or from a polyendocrine autoimmune disorder that is disrupting both thyroid and testicular function.

Gonadal Steroidogenesis and the Thyroid

Hypothyroidism in men has been shown to cause diminished libido, impotence, and oligospermia. It appears that hypothyroidism alters the metabolism of both androgens and estrogens. Although hypothyroidism is not as common in males, it must be ruled out with altered states of male hormone